

Full_Code_NLP_RAG_Project_Maneesh

May 25, 2025

0.1 Problem Statement

0.1.1 Business Context

The healthcare industry is rapidly evolving, with professionals facing increasing challenges in managing vast volumes of medical data while delivering accurate and timely diagnoses. The need for quick access to comprehensive, reliable, and up-to-date medical knowledge is critical for improving patient outcomes and ensuring informed decision-making in a fast-paced environment.

Healthcare professionals often encounter information overload, struggling to sift through extensive research and data to create accurate diagnoses and treatment plans. This challenge is amplified by the need for efficiency, particularly in emergencies, where time-sensitive decisions are vital. Furthermore, access to trusted, current medical information from renowned manuals and research papers is essential for maintaining high standards of care.

To address these challenges, healthcare centers can focus on integrating systems that streamline access to medical knowledge, provide tools to support quick decision-making, and enhance efficiency. Leveraging centralized knowledge platforms and ensuring healthcare providers have continuous access to reliable resources can significantly improve patient care and operational effectiveness.

Common Questions to Answer

1. **Diagnostic Assistance:** “What are the common symptoms and treatments for pulmonary embolism?”
2. **Drug Information:** “Can you provide the trade names of medications used for treating hypertension?”
3. **Treatment Plans:** “What are the first-line options and alternatives for managing rheumatoid arthritis?”
4. **Specialty Knowledge:** “What are the diagnostic steps for suspected endocrine disorders?”
5. **Critical Care Protocols:** “What is the protocol for managing sepsis in a critical care unit?”

0.1.2 Objective

As an AI specialist, your task is to develop a RAG-based AI solution using renowned medical manuals to address healthcare challenges. The objective is to **understand** issues like information overload, **apply** AI techniques to streamline decision-making, **analyze** its impact on diagnostics and patient outcomes, **evaluate** its potential to standardize care practices, and **create** a functional prototype demonstrating its feasibility and effectiveness.

0.1.3 Data Description

The **Merck Manuals** are medical references published by the American pharmaceutical company Merck & Co., that cover a wide range of medical topics, including disorders, tests, diagnoses, and drugs. The manuals have been published since 1899, when Merck & Co. was still a subsidiary of the German company Merck.

The manual is provided as a PDF with over 4,000 pages divided into 23 sections.

0.2 Installing and Importing Necessary Libraries and Dependencies

```
[1]: #Code to detect presence of GPU
import torch

if torch.cuda.is_available():
    device = torch.device("cuda")
    print("GPU is available.")
    print(f"Number of GPUs available: {torch.cuda.device_count()}")
    print(f"GPU Name: {torch.cuda.get_device_name(0)}")
    LLAMA_CUBLAS_OPTION = 'on'
    DEVICE_NAME = 'GPU'
    print(f"CUBLAS: {LLAMA_CUBLAS_OPTION}")
else:
    device = torch.device("cpu")
    print("GPU is not available, using CPU.")
    LLAMA_CUBLAS_OPTION = 'off'
    DEVICE_NAME = 'CPU'
    print(f"CUBLAS: {LLAMA_CUBLAS_OPTION}")

print(f"Device: {device}")
```

```
GPU is available.
Number of GPUs available: 1
GPU Name: Tesla T4
CUBLAS: on
Device: cuda
```

```
[2]: # Installation of llama-cpp-python (GPU/CPU)
#!CMAKE_ARGS="-DLLAMA_CUBLAS=LLAMA_CUBLAS_OPTION" FORCE_CMAKE=1 pip install_
↪ llama-cpp-python==0.1.85 --force-reinstall --no-cache-dir -q
```

```
[3]: # Library error avoidance
# Uninstall some current library versions
!pip uninstall -y numpy numba llama-cpp-python nvidia-cublas-cu12_
↪ nvidia-cuda-cupti-cu12 _
nvidia-cuda-nvrtc-cu12 nvidia-cuda-runtime-cu12 nvidia-cudnn-cu12_
↪ nvidia-cufft-cu12 nvidia-curand-cu12 \
nvidia-cusolver-cu12 nvidia-cuspars-cu12 nvidia-nvjitlink-cu12
```

```
# Install the required library versions
!CMAKE_ARGS="-DLLAMA_CUBLAS=LLAMA_CUBLAS_OPTION" FORCE_CMAKE=1 pip install
↳"numpy==2" "numba==0.60.0"
"llama-cpp-python==0.1.85" "nvidia-cublas-cu12==12.4.5.8"
↳"nvidia-cuda-cupti-cu12==12.4.127" \
"nvidia-cuda-nvrtc-cu12==12.4.127" "nvidia-cuda-runtime-cu12==12.4.127"
↳"nvidia-cudnn-cu12==9.1.0.70" \
"nvidia-cufft-cu12==11.2.1.3" "nvidia-curand-cu12==10.3.5.147"
↳"nvidia-cusolver-cu12==11.6.1.9" \
"nvidia-cuspars-cu12==12.3.1.170" "nvidia-nvjitlink-cu12==12.4.127" \
--force-reinstall --no-cache-dir -q
```

```
Found existing installation: numpy 2.0.2
Uninstalling numpy-2.0.2:
  Successfully uninstalled numpy-2.0.2
Found existing installation: numba 0.60.0
Uninstalling numba-0.60.0:
  Successfully uninstalled numba-0.60.0
WARNING: Skipping llama-cpp-python as it is not installed.

Found existing installation: nvidia-cublas-cu12 12.5.3.2
Uninstalling nvidia-cublas-cu12-12.5.3.2:
  Successfully uninstalled nvidia-cublas-cu12-12.5.3.2
Found existing installation: nvidia-cuda-cupti-cu12 12.5.82
Uninstalling nvidia-cuda-cupti-cu12-12.5.82:
  Successfully uninstalled nvidia-cuda-cupti-cu12-12.5.82
Found existing installation: nvidia-cuda-nvrtc-cu12 12.5.82
Uninstalling nvidia-cuda-nvrtc-cu12-12.5.82:
  Successfully uninstalled nvidia-cuda-nvrtc-cu12-12.5.82
Found existing installation: nvidia-cuda-runtime-cu12 12.5.82
Uninstalling nvidia-cuda-runtime-cu12-12.5.82:
  Successfully uninstalled nvidia-cuda-runtime-cu12-12.5.82
Found existing installation: nvidia-cudnn-cu12 9.3.0.75
Uninstalling nvidia-cudnn-cu12-9.3.0.75:
  Successfully uninstalled nvidia-cudnn-cu12-9.3.0.75
Found existing installation: nvidia-cufft-cu12 11.2.3.61
Uninstalling nvidia-cufft-cu12-11.2.3.61:
  Successfully uninstalled nvidia-cufft-cu12-11.2.3.61
Found existing installation: nvidia-curand-cu12 10.3.6.82
Uninstalling nvidia-curand-cu12-10.3.6.82:
  Successfully uninstalled nvidia-curand-cu12-10.3.6.82
Found existing installation: nvidia-cusolver-cu12 11.6.3.83
Uninstalling nvidia-cusolver-cu12-11.6.3.83:
  Successfully uninstalled nvidia-cusolver-cu12-11.6.3.83
Found existing installation: nvidia-cuspars-cu12 12.5.1.3
Uninstalling nvidia-cuspars-cu12-12.5.1.3:
  Successfully uninstalled nvidia-cuspars-cu12-12.5.1.3
Found existing installation: nvidia-nvjitlink-cu12 12.5.82
```

```

Uninstalling nvidia-nvjitlink-cu12-12.5.82:
  Successfully uninstalled nvidia-nvjitlink-cu12-12.5.82
                                60.9/60.9 kB
53.4 MB/s eta 0:00:00
                                1.8/1.8 MB
227.2 MB/s eta 0:00:00
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
                                19.3/19.3 MB
30.1 MB/s eta 0:00:00
                                3.7/3.7 MB
211.4 MB/s eta 0:00:00
                                363.4/363.4 MB
260.3 MB/s eta 0:00:00
                                13.8/13.8 MB
178.3 MB/s eta 0:00:00
                                24.6/24.6 MB
255.0 MB/s eta 0:00:00
                                883.7/883.7 kB
337.6 MB/s eta 0:00:00
                                664.8/664.8 MB
292.5 MB/s eta 0:00:00
                                211.5/211.5 MB
135.2 MB/s eta 0:00:00
                                56.3/56.3 MB
178.5 MB/s eta 0:00:00
                                127.9/127.9 MB
172.5 MB/s eta 0:00:00
                                207.5/207.5 MB
256.4 MB/s eta 0:00:00
                                21.1/21.1 MB
269.4 MB/s eta 0:00:00
                                45.5/45.5 kB
245.0 MB/s eta 0:00:00
                                43.9/43.9 MB
222.1 MB/s eta 0:00:00
                                45.8/45.8 kB
282.0 MB/s eta 0:00:00
  Building wheel for llama-cpp-python (pyproject.toml) ...
done

```

```

[4]: # For installing the libraries & downloading models from HF Hub
      #!pip install huggingface_hub==0.23.2 pandas==1.5.3 tiktoken==0.6.0 pymupdf==1.
      ↪25.1 langchain==0.1.1 \
      # langchain-community==0.0.13 chromadb==0.4.22 sentence-transformers==2.3.1
      ↪numpy==1.25.2 -q

```

```
!pip install huggingface_hub pandas tiktoken pymupdf langchain_
↳langchain-community chromadb sentence-transformers -q
```

```
67.3/67.3 kB
5.5 MB/s eta 0:00:00
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing metadata (pyproject.toml) ... done
24.1/24.1 MB
103.0 MB/s eta 0:00:00
2.5/2.5 MB
111.4 MB/s eta 0:00:00
19.0/19.0 MB
124.1 MB/s eta 0:00:00
94.9/94.9 kB
10.3 MB/s eta 0:00:00
284.2/284.2 kB
26.8 MB/s eta 0:00:00
2.0/2.0 MB
97.2 MB/s eta 0:00:00
101.6/101.6 kB
10.5 MB/s eta 0:00:00
16.4/16.4 MB
41.6 MB/s eta 0:00:00
65.8/65.8 kB
535.0 kB/s eta 0:00:00
55.9/55.9 kB
5.8 MB/s eta 0:00:00
194.9/194.9 kB
19.0 MB/s eta 0:00:00
119.0/119.0 kB
11.1 MB/s eta 0:00:00
96.7/96.7 kB
9.8 MB/s eta 0:00:00
44.4/44.4 kB
4.3 MB/s eta 0:00:00
62.5/62.5 kB
7.0 MB/s eta 0:00:00
459.8/459.8 kB
38.9 MB/s eta 0:00:00
50.9/50.9 kB
4.9 MB/s eta 0:00:00
71.5/71.5 kB
7.5 MB/s eta 0:00:00
4.0/4.0 MB
111.7 MB/s eta 0:00:00
454.8/454.8 kB
38.0 MB/s eta 0:00:00
```

```
46.0/46.0 kB
4.4 MB/s eta 0:00:00
86.8/86.8 kB
9.3 MB/s eta 0:00:00
Building wheel for pypika (pyproject.toml) ... done
```

```
[5]: #Libraries for processing dataframes, text
import json, os
import tiktoken
import pandas as pd
from datetime import datetime
import textwrap

#Libraries for Loading Data, Chunking, Embedding, and Vector Databases
from langchain.text_splitter import RecursiveCharacterTextSplitter
from langchain_community.document_loaders import PyMuPDFLoader
from langchain_community.embeddings.sentence_transformer import SentenceTransformerEmbeddings
from langchain_community.vectorstores import Chroma

#Libraries for downloading and loading the llm
from huggingface_hub import hf_hub_download
from llama_cpp import Llama
```

0.3 Question Answering using LLM

Downloading and Loading the model

```
[6]: # Hugging face repository and model version
model_name_or_path = "TheBloke/Mistral-7B-Instruct-v0.2-GGUF"

# Quantized mistral model in GGUF format
model_basename = "mistral-7b-instruct-v0.2.Q6_K.gguf"

[7]: model_path = hf_hub_download(
    repo_id=model_name_or_path,
    filename=model_basename
)
```

```
/usr/local/lib/python3.11/dist-packages/huggingface_hub/utils/_auth.py:94:
UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab
(https://huggingface.co/settings/tokens), set it as secret in your Google Colab
and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access
public models or datasets.
```

```
warnings.warn(
mistral-7b-instruct-v0.2.Q6_K.gguf: 0%|          | 0.00/5.94G [00:00<?, ?B/s]
```

```
[8]: if DEVICE_NAME == 'GPU': #For GPU
    llm = Llama(
        model_path=model_path,
        n_ctx=2300,
        n_gpu_layers=38,
        n_batch=512
    )
else: #For CPU
    llm = Llama(
        model_path=model_path,
        n_ctx=1024
    )
```

```
AVX = 1 | AVX2 = 1 | AVX512 = 0 | AVX512_VBMI = 0 | AVX512_VNNI = 0 | FMA = 1 |
NEON = 0 | ARM_FMA = 0 | F16C = 1 | FP16_VA = 0 | WASM_SIMD = 0 | BLAS = 1 |
SSE3 = 1 | SSSE3 = 1 | VSX = 0 |
```

Response

```
[9]: def response(query,max_tokens=512,temperature=0,top_p=0.95,top_k=50):
    model_output = llm(
        prompt=query,
        max_tokens=max_tokens,
        temperature=temperature,
        top_p=top_p,
        top_k=top_k
    )

    return model_output['choices'][0]['text'].strip() #Extract the text portions
```

0.3.1 Query 1: What is the protocol for managing sepsis in a critical care unit?

```
[10]: user_query_1 = "What is the protocol for managing sepsis in a critical care_
    ↪unit?"

start_time = datetime.now()

response_string_1 = textwrap.fill(response(user_query_1), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
```

```
print(response_string_1)
```

Time taken for query response: 20.44923 seconds

Sepsis is a life-threatening condition that can arise from an infection, and it requires prompt recognition and

aggressive management in a critical care unit. The following are general steps for managing sepsis in a critical care

unit: 1. Early recognition: Recognize the signs and symptoms of sepsis early and initiate treatment as soon as

possible. Sepsis can present with various clinical features, including fever or hypothermia, tachycardia or bradycardia,

altered mental status, respiratory distress, and lactic acidosis. 2.

Resuscitation: Provide adequate fluid resuscitation

to maintain adequate tissue perfusion. The goal is to achieve a mean arterial pressure (MAP) of at least 65 mmHg and a

central venous oxygen saturation (ScvO2) of greater than 70%. 3. Antibiotics:

Administer broad-spectrum antibiotics as

soon as possible based on the suspected source of infection and local

microbiology data. 4. Source control: Identify and

address the source of infection, if possible. This may involve surgical intervention or other interventions such as

drainage or debridement. 5. Vasopressors: If adequate fluid resuscitation does not maintain adequate tissue perfusion,

consider the use of vasopressors to maintain MAP. 6. Corticosteroids: Consider the use of corticosteroids in patients

with sepsis who are not responding to initial resuscitative efforts. 7.

Inotropes: Consider the use of inotropic agents

to improve cardiac output and tissue perfusion. 8. Renal replacement therapy:

Consider the use of renal replacement

therapy in patients with sepsis who have acute kidney injury or who are not responding to other interventions. 9. Fluid

management: Carefully manage fluid balance to avoid overload while maintaining adequate tissue perfusion. 10.

Monitoring: Close monitoring of vital signs, laboratory values, and clinical status is essential for effective sepsis

management in a critical care unit. 11. Early goal-directed therapy (EGDT): EGDT is a protocolized approach to sepsis

management that involves early recognition, aggressive fluid resuscitation, and goal-directed interventions to optimize

tissue perfusion. It has been shown to improve outcomes in patients with

- The response generated by the model is quite appropriate. (When compared to a google gemini or web search)

0.3.2 Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
[11]: user_query_2 = "What are the common symptoms for appendicitis, and can it be_
      ↪cured via medicine? If not, what surgical procedure should be followed to_
      ↪treat it?"

      start_time = datetime.now()

      response_string_2 = textwrap.fill(response(user_query_2), width=120)

      end_time = datetime.now()

      elapsed_time = end_time - start_time

      print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")

      print(response_string_2)
```

Llama.generate: prefix-match hit

Time taken for query response: 20.810534 seconds

Appendicitis is a medical condition characterized by inflammation of the appendix, a small tube-shaped organ located in the lower right side of the abdomen. The symptoms of appendicitis can vary from person to person, but some common signs include:

1. Abdominal pain: The pain may start as a mild discomfort around the navel or in the lower right abdomen, which then gradually moves to the right lower quadrant and becomes more severe over time. The pain may be constant or intermittent and is often worsened by movement, coughing, or deep breathing.
2. Loss of appetite: People with appendicitis may lose their appetite due to abdominal pain and discomfort.
3. Nausea and vomiting: Vomiting is a common symptom of appendicitis, especially in the later stages of the condition.
4. Fever: A fever of 100.4°F (38°C) or higher may be present in some cases of appendicitis.
5. Constipation or diarrhea: Both constipation and diarrhea can occur with appendicitis, depending on the location and severity of the inflammation.
6. Rebound tenderness: This is a sign of inflammation in the abdomen. When pressure is applied to the abdomen, the pain worsens when the pressure is released.
7. Guarding: People with appendicitis may involuntarily contract their abdominal muscles to protect the affected area from further discomfort or injury.

Appendicitis cannot be cured via medicine alone. Antibiotics may be used to treat any accompanying bacterial infection, but they do not address the underlying inflammation in the appendix. Surgery is the

only effective treatment for appendicitis. The most common surgical procedure for treating appendicitis is an appendectomy, which involves removing the inflamed appendix through a small incision in the abdomen. This procedure can be performed as an open surgery or laparoscopically (using small incisions and a camera to guide the surgeon). The choice of surgical approach depends on various factors, including the severity of the condition, the patient's overall health, and the surgeon's preference. It is essential to seek medical attention promptly if you suspect appendicitis,

- Similar to the first response, the model is performing well.

0.3.3 Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
[12]: user_query_3 = "What are the effective treatments or solutions for addressing_
      ↪sudden patchy hair loss, commonly seen as localized bald spots on the scalp,_
      ↪and what could be the possible causes behind it?"

start_time = datetime.now()

response_string_3 = textwrap.fill(response(user_query_3), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")

print(response_string_3)
```

Llama.generate: prefix-match hit

Time taken for query response: 21.082282 seconds

Sudden patchy hair loss, also known as alopecia areata, is a common autoimmune disorder that affects the hair follicles.

It can result in round or oval bald patches on the scalp, but it can also occur on other parts of the body such as the

beard area, eyebrows, or eyelashes. The exact cause of alopecia areata is not known, but it's believed to be related to

a problem with the immune system. Some possible triggers for this condition include stress, genetics, viral infections,

and certain medications. There are several treatments that have been shown to be effective in addressing sudden patchy

hair loss: 1. Corticosteroids: These are anti-inflammatory drugs that can help reduce inflammation and suppress the

immune system's attack on the hair follicles. They can be applied topically or

taken orally, depending on the severity of the condition. 2. Minoxidil: This is a medication that has been shown to promote hair growth in some people with alopecia areata. It works by increasing blood flow to the affected area and stimulating hair follicles to produce new hairs. 3. DHT blockers: Dihydrotestosterone (DHT) is a hormone that can contribute to hair loss. Some people with alopecia areata have high levels of DHT in their scalp. Finasteride and dutasteride are medications that can block the production of DHT, which may help promote hair growth. 4. Immune system modulators: These drugs work by suppressing the immune system's attack on the hair follicles. Examples include methotrexate, mycophenolate mofetil, and cyclosporine. 5. Hair transplantation: In some cases, hair transplantation may be an option for people with alopecia areata who have large bald patches that don't respond to other treatments. This involves taking healthy hairs from another part of the scalp or body and transplanting them into the affected area. It's important to note that there is no cure for alopecia areata, and not all treatments work for everyone. The effectiveness of treatment can also depend on the severity and location of the bald patches. If you're experiencing sudden patchy

- The model seems to be working well with common solutions provided as the response.

0.3.4 Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
[13]: user_query_4 = "What treatments are recommended for a person who has sustained_
      ↪a physical injury to brain tissue, resulting in temporary or permanent_
      ↪impairment of brain function?"

start_time = datetime.now()

response_string_4 = textwrap.fill(response(user_query_4), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_4)
```

Llama.generate: prefix-match hit

Time taken for query response: 19.42717 seconds

A person who has sustained a physical injury to brain tissue, also known as a traumatic brain injury (TBI), may require

various treatments depending on the severity and location of the injury. Here are some common treatments recommended for

TBIs: 1. Emergency care: The first priority is to ensure the person's airway is clear, they are breathing, and their heart is beating normally. In severe cases, emergency surgery may be required to remove hematomas or other obstructions.

2. Medications: Depending on the symptoms, medications may be prescribed to manage conditions such as swelling, seizures, pain, or infections. For example, corticosteroids may be used to reduce brain swelling, and anticonvulsants

may be given to prevent seizures. 3. Rehabilitation: Rehabilitation is an essential part of the recovery process for TBI

patients. Rehabilitation may include physical therapy to help with mobility and strength, occupational therapy to help with daily living skills, speech therapy to improve communication, and cognitive rehabilitation to address memory,

attention, and other cognitive impairments. 4. Assistive devices: Depending on the extent of the injury, assistive

devices such as wheelchairs, walkers, or braces may be necessary to help with mobility and independence. 5. Surgery: In

some cases, surgery may be required to remove hematomas or other obstructions that are putting pressure on the brain.

Craniotomies, which involve removing a portion of the skull to relieve pressure on the brain, may also be necessary in

severe cases. 6. Nutritional support: TBI patients may require nutritional support through feeding tubes or intravenous

nutrition if they are unable to eat or drink normally. 7. Psychological support: TBI patients and their families may

benefit from psychological support to help cope with the emotional and psychological challenges of recovery. This may

include counseling, support groups, or other forms of therapy. 8. Lifestyle modifications: Depending on the extent of

the injury, lifestyle modifications such as avoiding alcohol, quitting smoking, and getting enough rest may be necessary

to promote healing and prevent further damage to the brain.

- The model seems to be working well with accurate reference to handling TBI.

0.3.5 Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
[14]: user_query_5 = "What are the necessary precautions and treatment steps for a_
      ↪person who has fractured their leg during a hiking trip, and what should be_
      ↪considered for their care and recovery?"
```

```
start_time = datetime.now()
```

```
response_string_5 = textwrap.fill(response(user_query_5), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_5)
```

Llama.generate: prefix-match hit

Time taken for query response: 19.543151 seconds

First and foremost, if you suspect that someone has fractured their leg while hiking, it's essential to ensure their safety and prevent further injury. Here are some necessary precautions:

1. Keep the person calm and still: Encourage them to remain as still as possible to minimize pain and prevent worsening the injury.
2. Assess the situation: Check for any signs of shock, such as pale skin, rapid heartbeat, or shallow breathing. If you notice these symptoms, seek medical help immediately.
3. Immobilize the leg: Use a splint, sling, or other available materials to immobilize the leg and prevent movement. Be sure not to apply too much pressure on the injury site.
4. Provide pain relief: Offer over-the-counter pain medication, such as acetaminophen or ibuprofen, to help manage pain.
5. Seek medical attention: If the fracture is severe or if you suspect that there may be other injuries, seek medical help as soon as possible.

Once you've ensured the person's safety and stability, consider the following steps for their care and recovery:

1. Transportation: Arrange for transportation to a hospital or medical facility as soon as possible. If it's not feasible to move the person, call emergency services for assistance.
2. Diagnosis and treatment: A healthcare professional will diagnose the fracture and determine the best course of treatment based on the severity and location of the injury. This may include immobilization with a cast or brace, surgery, or other interventions.
3. Rehabilitation: After the initial treatment, the person will need to undergo rehabilitation to regain strength, flexibility, and mobility in the affected leg. This may involve physical therapy exercises, assistive devices, and other interventions.
4. Follow-up care: Regular follow-up appointments with a healthcare professional are essential to monitor the healing process and ensure that the fracture is healing correctly.
5. Prevention: Encourage the person to take steps to prevent future injuries while hiking, such as wearing appropriate footwear, using trekking poles, and staying on marked trails.

- The model seems to be performing well and provides appropriate steps to handling a leg fracture.

0.3.6 Observations on using the LLM

- Responses are presented in a logical, easy-to-follow structure— using numbered lists—to enhance readability and comprehension.
- The model effectively interprets and responds to user questions with information that is contextually appropriate, ensuring key aspects of the query are addressed.
- Many outputs are cut off mid-sentence (e.g., explanations on treatments for traumatic brain injury or fractures), most likely due to token constraints. This hampers the completeness of the information and may frustrate users seeking complete answers.
- The responses are observed to be generic with no specialized information generated or retrieved. This can make responses feel like general guidelines rather than nuanced, tailored insights suited to complex or specialized medical situations.

0.4 Question Answering using LLM with Prompt Engineering

```
[15]: #Define function for response with prompt engineering (PE)
def response_with_PE(instruction, query,max_tokens=512,temperature=0,top_p=0.
    ↪95,top_k=50, repeat_penalty = 1.2):
    # System message
    system_message = """
        [INST]<<SYS>>
        {}
        <</SYS>>[/INST]
        """.format(instruction)

    prompt = f"{system_message}\n{query}"

    #Generate the response
    try:
        model_output = llm(
            prompt=prompt,
            max_tokens=max_tokens,
            temperature=temperature,
            top_p=top_p,
            top_k=top_k,
            repeat_penalty= repeat_penalty,
            stop=['INST'],
            echo=False
        )
        response = model_output['choices'][0]['text'].strip()
    except Exception as e:
        response = f'Sorry, I encountered the following error: \n {e}'

    return response
```

0.4.1 Query 1: What is the protocol for managing sepsis in a critical care unit?

```
[16]: instruction_string = "Playing the role of a medical information specialist,␣  
      ↪answer the user query with clarity and brevity."  
  
[17]: start_time = datetime.now()  
  
      response_string_PE_1 = textwrap.fill(response_with_PE(instruction_string,␣  
      ↪user_query_1,temperature = 0), width=120)  
  
      end_time = datetime.now()  
  
      elapsed_time = end_time - start_time  
  
      print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")  
      print(response_string_PE_1)
```

Llama.generate: prefix-match hit

Time taken for query response: 11.286336 seconds

Sepsis is a life-threatening condition caused by the body's response to infection. The following steps are typically taken in a critical care setting: 1. Early recognition and suspicion of sepsis based on clinical signs, such as fever or hypothermia, tachycardia or bradycardia, respiratory distress, altered mental status, and lactic acidosis. 2. Immediate initiation of broad-spectrum antibiotics to cover potential pathogens while awaiting culture results. 3. Aggressive fluid resuscitation with intravenous crystalloids to maintain adequate tissue perfusion and organ function. 4. Close monitoring of vital signs, oxygenation, and hemodynamic status. 5. Adjustment of antibiotics based on culture and sensitivity results. 6. Administration of vasopressors or inotropes if needed for blood pressure support. 7. Use of corticosteroids, anticoagulation, or other adjunctive therapies as per clinical guidelines and individual patient response. 8. Supportive care measures such as mechanical ventilation, renal replacement therapy, and nutritional support as required. 9. Continuous monitoring for signs of sepsis resolution or progression to septic shock.

```
[18]: # Change the temperature to 1 and rerun the prompt  
      start_time = datetime.now()  
  
      response_string_PE_1 = textwrap.fill(response_with_PE(instruction_string,␣  
      ↪user_query_1, temperature = 1), width=120)  
  
      end_time = datetime.now()
```

```

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_PE_1)

```

Llama.generate: prefix-match hit

Time taken for query response: 20.17796 seconds

1. Early recognition: Look out for signs of infection such as fever, chills, tachycardia, respiratory distress, or altered mental status. Use the Sequential Organ Failure Assessment (SOFA) score to assess severity. 2. Fluid resuscitation: Administer intravenous crystalloid solution in adequate amounts until endpoints are met - this includes maintaining mean arterial pressure >65mmHg, urine output 0.5mL/kg/h and adequate tissue perfusion as evidenced by improved organ function. 3. Antibiotics: Begin broad-spectrum antibiotic therapy promptly based on culture sensitivities when available and continue for at least 48 hours beyond defervescence or clinical improvement. 4. Source control: Address underlying infection source, such as drainage of abscesses, removal of necrotic tissue or surgical intervention if necessary. 5. Vasopressors: Use vasopressors to maintain adequate blood pressure when fluid resuscitation is insufficient and sepsis-induced hypotension persists. 6. Inotropes: Administer inotropic agents if needed for cardiac support, but minimize dosage as much as possible due to the potential adverse effects on other organs. 7. Corticosteroids: Consider low-dose steroid use only when there is evidence of refractory shock or ARDS unresponsive to other treatments; this should be under the guidance and supervision of a critical care specialist. 8. Monitoring and supportive measures: Provide regular monitoring for vital signs, laboratory tests including lactate levels, blood gases, coagulation status, electrolytes, urine output, cardiac function, and oxygenation requirements; optimize fluid balance by restricting excess fluids to prevent pulmonary edema. 9. Mechanical ventilation: Use mechanical ventilation as needed for respiratory support in cases of severe sepsis or septic shock with underlying ARDS. 10. Adjunctive treatments: Consider other therapies, such as anticoagulation therapy and continuous renal replacement therapy (CRRT), if the patient's condition warrants it based on clinical judgement and expert consultation.

Impact of Temperature: * **Temperature = 0:** The response is concise and factual, reflecting common knowledge.

- **Temperature = 1:** Interestingly, the output for temperature=1 is identical to temperature=0 for the non-RAG query.
- Possible Reasons for Identical Output: For a very common and well-defined medical query like this, the LLM might have a very strong “default” answer that even a higher temperature doesn’t significantly deviate from with such a simple prompt.
- The max_length=512 might be constraining, though the answer is short.
- It is possible that for this specific input and model, the probability distribution for the next tokens is sharply peaked, leading to the same output even with increased sampling randomness.
- While not evident here, typically temperature=1 would introduce more variability. For factual Q&A from a specific document, this variability is usually undesirable as it can lead to answers not supported by the source text.

0.4.2 Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
[19]: start_time = datetime.now()

response_string_PE_2 = textwrap.fill(response_with_PE(instruction_string,
↪user_query_2, top_p = 0.95), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_PE_2)
```

Llama.generate: prefix-match hit

Time taken for query response: 6.888298 seconds

Answer: Appendicitis is characterized by abdominal pain, usually localized in the lower right side of the belly. Other common symptoms include loss of appetite, nausea, vomiting, fever, and a feeling of being sick or unwell. Appendicitis cannot be cured with medicine alone; it typically requires surgical removal of the appendix (appendectomy) to prevent rupture and potential complications such as peritonitis. The most common type of appendectomy is laparoscopic, which uses small incisions and a camera for minimally invasive surgery. Open appendectomies are less commonly performed but may be necessary in certain cases. After the procedure, patients usually recover within 1-2 weeks with proper care and follow-up.

```
[77]: # Change the top_p to 2 and rerun the prompt
start_time = datetime.now()
```

```

response_string_PE_2 = textwrap.fill(response_with_PE(instruction_string,
↳user_query_2,top_p = 2.0), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_PE_2)

```

Llama.generate: prefix-match hit

Time taken for query response: 7.008051 seconds

Answer: Appendicitis is characterized by abdominal pain, usually localized in the lower right side of the belly. Other common symptoms include loss of appetite, nausea, vomiting, fever, and a feeling of being sick or unwell. Appendicitis cannot be cured with medicine alone; it typically requires surgical removal of the appendix (appendectomy) to prevent rupture and potential complications such as peritonitis. The most common type of appendectomy is laparoscopic, which uses small incisions and a camera for minimally invasive surgery. Open appendectomies are less commonly performed but may be necessary in certain cases. After the procedure, patients usually recover within 1-2 weeks with proper care and follow-up.

- A **top_p** value is typically between 0 and 1. 0 is excluded and 1 is included.
- The value used here for top_p (2) is outside the range and probably replaced with 1. This could be the reason that the responses are very similar.

0.4.3 Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```

[21]: start_time = datetime.now()

response_string_PE_3 = textwrap.fill(response_with_PE(instruction_string,
↳user_query_3), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_PE_3)

```

Llama.generate: prefix-match hit

Time taken for query response: 5.3664 seconds

Answer: Sudden patchy hair loss can have various underlying causes such as alopecia areata (an autoimmune condition), nutritional deficiencies, stress, or certain medications. Treatment options depend on the cause but may include topical corticosteroids for inflammatory conditions like alopecia areata, dietary modifications for nutrient-related causes, and addressing underlying emotional or physical stressors. In severe cases, hair transplantation might be considered as a last resort. Consulting with a healthcare professional is recommended to determine the cause and appropriate treatment plan.

```
[22]: # Change the top_k to 30 and rerun the prompt
start_time = datetime.now()

response_string_PE_3 = textwrap.fill(response_with_PE(instruction_string,
↪user_query_3, top_k = 30), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_PE_3)
```

Llama.generate: prefix-match hit

Time taken for query response: 5.209535 seconds

Answer: Sudden patchy hair loss can have various underlying causes such as alopecia areata (an autoimmune condition), nutritional deficiencies, stress, or certain medications. Treatment options depend on the cause but may include topical corticosteroids for inflammatory conditions like alopecia areata, dietary modifications for nutrient-related causes, and addressing underlying emotional or physical stressors. In severe cases, hair transplantation might be considered as a last resort. Consulting with a healthcare professional is recommended to determine the cause and appropriate treatment plan.

- **top_k** sampling is expected to restrict the model's choice for the next k probable tokens. Smaller k is supposed to make the output more focussed and less random.
- In our case here, we used k = 50 and then k = 30. Though they are large numbers, the responses are very similar as k = 30 will steer the answer towards most probable sequence. The model is confident of its standard answer.
- Maybe we need to use more restrictive values closer to 1 or 2.

0.4.4 Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
[23]: start_time = datetime.now()

response_string_PE_4 = textwrap.fill(response_with_PE(instruction_string,␣
↪user_query_4, repeat_penalty = 1.2), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_PE_4)
```

Llama.generate: prefix-match hit

Time taken for query response: 8.012691 seconds

Answer: The treatment for traumatic brain injuries (TBIs) depends on the severity and location of the damage. For mild TBIs, rest, hydration, and over-the-counter pain relievers may be sufficient. More severe cases might require hospitalization, surgery, or rehabilitation therapies such as physical therapy, occupational therapy, speech therapy, and cognitive rehabilitation to help restore lost skills and improve overall function. Medications like diuretics, sedatives, anticonvulsants, and anti-inflammatory drugs may also be prescribed to manage symptoms or prevent complications. In some cases, assistive devices such as wheelchairs, braces, or communication aids might be necessary for long-term care. It's essential that individuals with TBIs receive proper medical attention and follow their healthcare provider's recommendations for the best possible outcome.

```
[24]: # Change repeat penalty to 5 and rerun the prompt
start_time = datetime.now()

response_string_PE_4 = textwrap.fill(response_with_PE(instruction_string,␣
↪user_query_4, repeat_penalty = 5), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_PE_4)
```

Llama.generate: prefix-match hit

Time taken for query response: 20.077413 seconds

The treatment options depend on various factors such as severeness of the injury and its location in the brain. Generally speaking: 1.) Immediate care includes managing airway and breathing with oxygen therapy if necessary; controlling bleeding through surgery (if there's a skull fracture or hematoma); preventing infection, etc., known collectively as a traumatic brain injury (TBI). 2.) Rehabilitation programs may include physical therapy to help regain mobility and strength in affected limbs. Speech-language therapy can assist with communication and swallowing issues if present due to damage in the speech or language areas of the brain tissue or nerve pathways connecting ear to mouth/larynx (voice box). Occupational therapy can help individuals learn new skills or tasks, modify their environment to accommodate their disability and improve their own independence. 3.) Medications may be prescribed to manage symptoms such as pain or seizures; improve cognitive function (memory and attention); reduce anxiety/depression etc., depending on the specific injury's symptoms and situation of patient (either short-term or long-term treatment). 4.) In severe cases, surgical intervention might be required to remove blood clots or repair damaged brain tissue. 5.) Alternative therapies like acupuncture and transcranial magnetic stimulation (TMS), though not universally accepted yet due to varying effectiveness across individuals with TBI injuries are sometimes used in conjunction with conventional therapy treatment plans for improving cognitive functioning and motor control. 6.) Support groups can provide emotional and social support for patients dealing with the challenges of living with a head injury, helping them cope better through shared experiences with others in the same position as well as stimulating positivity among members by encouraging mutual understanding and acceptance. 7.) Lastly but importantly: family and friends play a crucial role in the recovery process by providing emotional support during this challenging time, encouraging participation in rehabilitation programs and everyday activities that help keep patients engaged and socialized while they heal from their brain trauma injury (TBI).

- Effect of **repeat_penalty** = 1.2 is a concise, high level summary covering common symptoms and broad treatment categories.
- Effect of **repeat_penalty** = 5 is a more detailed, lengthy summary.
- A mild penalty (1.2) allows the model to generate a natural but balanced response.
- A high penalty is probably forcing the model to avoid repetition and phrases. This also is the reason for long dense sentences seen. (For example: severeness of the injury and its location in the brain)
- Care needs to be taken while using repeat_penalty values.

0.4.5 Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
[25]: start_time = datetime.now()

response_string_PE_5 = textwrap.fill(response_with_PE(instruction_string,
↳user_query_5, max_tokens = 512), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_PE_5)
```

Llama.generate: prefix-match hit

Time taken for query response: 9.165604 seconds

Answer: A fractured leg requires immediate medical attention. The following steps can help manage the situation until professional medical assistance arrives: 1. Immobilize the affected limb using a splint or sling to prevent further damage and reduce pain. 2. Apply ice packs for 15-20 minutes at a time, several times a day, to minimize swelling and inflammation. 3. Keep the injured leg elevated above heart level whenever possible to help decrease swelling and discomfort. 4. Provide adequate hydration and nutrition to support healing. 5. Monitor for signs of infection such as redness, warmth, or increased pain around the fracture site. 6. Seek medical care promptly upon reaching civilization. X-rays will be needed to confirm the diagnosis and determine the extent of the injury. Depending on the severity of the fracture, treatment may include immobilization with a cast or brace, surgery, or physical therapy. Proper follow-up care is essential for complete recovery.

```
[26]: # Change max_tokens to 1024 and rerun the prompt

start_time = datetime.now()

response_string_PE_5 = textwrap.fill(response_with_PE(instruction_string,
↳user_query_5, max_tokens = 1024), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time
```

```
print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_PE_5)
```

Llama.generate: prefix-match hit

Time taken for query response: 10.778962 seconds

Answer: A fractured leg requires immediate medical attention. The following steps can help ensure proper care and recovery: 1. Immobilize the affected limb using a splint or cast to prevent further damage and promote healing. 2. Apply ice packs intermittently for 15-20 minutes at a time, several times a day, to reduce swelling and pain. 3. Elevate the leg above heart level whenever possible to minimize swelling and discomfort. 4. Monitor and manage pain with over-the-counter or prescription medications as recommended by a healthcare professional. 5. Keep the wound clean and dry to prevent infection; cover it with a sterile dressing if necessary. 6. Follow a doctor's prescribed rehabilitation plan, which may include physical therapy exercises and assistive devices like crutches or a walker. 7. Maintain good nutrition and hydration for optimal healing and overall health. 8. Avoid putting weight on the injured leg until it has healed completely to prevent complications such as delayed union or non-union fractures. 9. Regularly attend follow-up appointments with a healthcare professional to monitor progress, address any concerns, and adjust treatment plans if necessary.

- With `max_length = 512`, the LLM provided a useful but condensed set of instructions in the response.
- With `max_length = 1024`, the LLM leveraged a larger token limit and provided a slightly more detailed recovery/treatment steps.
- Both responses were common sense advice for managing a fracture.

0.4.6 Observations on using LLM with Prompt Engineering

- Prompt engineered queries are seen to be more faster, yielding more specific and relevant responses above.
- It provides more control over the format and style of the responses.
- Provides more consistent answers (I experimented thrice on the responses)
- Individual combination observations are noted below the queries.

0.5 Data Preparation for RAG

0.5.1 Loading the Data

```
[28]: # This is specific to my google colab environment
from google.colab import drive
```

```
drive.mount('/content/drive',force_remount = True)

FOLDER_PATH = '/content/drive/MyDrive/Colab_Notebooks/Medical_Assistant_Project/
↳ '

medical_manual_path = FOLDER_PATH+"medical_diagnosis_manual.pdf"
```

Mounted at /content/drive

```
[29]: #Load the pdf
pdf_loader = PyMuPDFLoader(medical_manual_path)
medical_manual = pdf_loader.load()
```

0.5.2 Data Overview

Checking the first 5 pages

```
[30]: #Print the first 5 pages of the pdf
for page in medical_manual[:5]:
    print(page.page_content)
```

```
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Chapter 116. Myeloproliferative Disorders	...
...	
..	
1141	
Chapter 117. Leukemias	...
...	
...	
1154	
Chapter 118. Lymphomas	...
...	
...	
1164	
Chapter 119. Plasma Cell Disorders	...
...	
...	
1172	

Chapter 120. Iron Overload	...
...	
...	
1177	
Chapter 121. Transfusion Medicine	...
...	
...	
1186	
Chapter 122. Overview of Cancer	...
...	
...	
1198	
Chapter 123. Tumor Immunology	...
...	
...	
1204	
Chapter 124. Principles of Cancer Therapy	...
...	
1215	
10 - Immunology; Allergic Disorders	...
...	
..	
1215	
Chapter 125. Biology of the Immune System	...
...	
1227	
Chapter 126. Immunodeficiency Disorders	...
...	
1243	
Chapter 127. Allergic & Other Hypersensitivity Disorders	...
...	
1263	
Chapter 128. Transplantation	...
...	
...	
1281	
11 - Infectious Diseases	...
...	
...	
1281	
Chapter 129. Biology of Infectious Disease	...
...	
.	
1300	
Chapter 130. Laboratory Diagnosis of Infectious Disease	...
...	
1306	
Chapter 131. Immunization	...

...	
...	
1313	
Chapter 132. Bacteria & Antibacterial Drugs	...
...	
1353	
Chapter 133. Gram-Positive Cocci	...
...	
...	
1366	
Chapter 134. Gram-Positive Bacilli	...
...	
...	
1376	
Chapter 135. Gram-Negative Bacilli	...
...	
...	
1405	
Chapter 136. Spirochetes	...
...	
...	
1413	
Chapter 137. Neisseriaceae	...
...	
...	
1419	
Chapter 138. Chlamydia & Mycoplasmas	...
...	
1421	
Chapter 139. Rickettsiae & Related Organisms	...
...	
1431	
Chapter 140. Anaerobic Bacteria	...
...	
...	
1450	
Chapter 141. Mycobacteria	...
...	
...	
1470	
Chapter 142. Fungi	...
...	
...	
1493	
Chapter 143. Approach to Parasitic Infections	...
...	
1496	
Chapter 144. Nematodes (Roundworms)	...

...
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Checking the number of pages

```
[31]: #Count the total number of pages
print(f"Total number of pages: {len(medical_manual)}")
```

Total number of pages: 4114

0.5.3 Data Chunking

```
[32]: # Data Chunking for splitting into more manageable chunks
text_splitter = RecursiveCharacterTextSplitter.from_tiktoken_encoder(
    encoding_name='cl100k_base',
    chunk_size=512,
    chunk_overlap= 50 #Increased the chunk_overlap from 20 to 50
)
```

- The experiment with `chunk_overlap = 20` did not yield much overlap and hence I changed the `chunk_overlap` to 50.

```
[33]: document_chunks = pdf_loader.load_and_split(text_splitter)
```

```
[34]: #Count the total number of chunks that the pdf has been split into
print(f"Total number of text chunks: {len(document_chunks)}")
```

Total number of text chunks: 8679

```
[35]: print(textwrap.fill(document_chunks[4000].page_content, width=120))
```

triptans and other vasoconstrictors. Chronic migraines: The same drugs used to prevent episodic migraine are used to treat chronic migraine. Prevention Daily preventive therapy is warranted when frequent migraines interfere with activity despite acute treatment. For patients who use analgesics frequently, particularly those with medication overuse headache, preventive drugs (see Table 178-4) should be combined with a program for stopping overused analgesics. Choice of drug can be guided by coexisting disorders, as for the following: • A small bedtime dose of amitriptyline for patients with insomnia • A α -blocker for patients with anxiety or coronary artery disease • Topiramate, which can induce weight loss, for obese patients or for patients who wish to avoid weight gain • Divalproex for patients with mania Post-Lumbar Puncture and Other Low-Pressure Headaches Low-pressure headaches result from reduction in CSF volume and pressure

due to lumbar puncture or spontaneous or traumatic CSF leaks. Removal of CSF by lumbar puncture (LP) reduces CSF volume and pressure, as do spontaneous or traumatic CSF leaks. Headache after LP is common, usually occurring hours to a day or two afterward, and can be severe. Younger patients with a small body mass are at greatest risk. Using small, noncutting needles reduces risk. The amount of CSF removed and duration of recumbency after LP do not affect incidence. Spontaneous CSF leaks may result when an arachnoid cyst along the spinal canal ruptures. Coughing or sneezing may cause the rupture. CSF may leak after certain head or facial injuries (eg, basilar skull fractures). Headache results when head elevation while sitting or standing stretches the pain-sensitive basal meninges. Headaches are intense, postural, and often accompanied by neck pain, meningismus, and vomiting. Headache is alleviated only by lying completely flat. Diagnosis • Clinical evaluation Post-LP headache is clinically obvious, and testing is rarely needed; other low-pressure headaches may require brain imaging. MRI with gadolinium often shows diffuse enhancement of the pachymeninges and,

```
[36]: print(textwrap.fill(document_chunks[4001].page_content, width=120))
```

Diagnosis • Clinical evaluation Post-LP headache is clinically obvious, and testing is rarely needed; other low-pressure headaches may require brain imaging. MRI with gadolinium often shows diffuse enhancement of the pachymeninges and, in severe cases, downward sagging of the brain. CSF pressure is typically low or unobtainable if patients have been upright for any length of time (gravity accelerates CSF loss). Treatment • Hydration and analgesics • Sometimes an epidural blood patch The Merck Manual of Diagnosis & Therapy, 19th Edition Chapter 178. Headache 1889 manandkn2016@gmail.com ZGIQC09SUD This file is meant for personal use by manandkn2016@gmail.com only. Sharing or publishing the contents in part or full is liable for legal action.

- As expected we notice some overlaps in the extracted text below:
 - Clinical evaluation Post-LP headache is clinically obvious, and testing is rarely needed; other low-pressure headaches may require brain imaging. MRI with gadolinium often shows diffuse enhancement of the pachymeninges and,

0.5.4 Embedding

```
[37]: embedding_model = SentenceTransformerEmbeddings(model_name='thenlper/gte-large')
```

```
<ipython-input-37-8acaa307165f>:1: LangChainDeprecationWarning: The class
`HuggingFaceEmbeddings` was deprecated in LangChain 0.2.2 and will be removed in
1.0. An updated version of the class exists in the :class:`~langchain-
```

huggingface package and should be used instead. To use it run ``pip install -U :class:`~langchain-huggingface`` and import as ``from :class:`~langchain_huggingface import HuggingFaceEmbeddings```.

```
embedding_model = SentenceTransformerEmbeddings(model_name='thenlper/gte-large')
```

```
modules.json: 0%|          | 0.00/385 [00:00<?, ?B/s]
README.md: 0%|          | 0.00/67.9k [00:00<?, ?B/s]
sentence_bert_config.json: 0%|          | 0.00/57.0 [00:00<?, ?B/s]
config.json: 0%|          | 0.00/619 [00:00<?, ?B/s]
model.safetensors: 0%|          | 0.00/670M [00:00<?, ?B/s]
tokenizer_config.json: 0%|          | 0.00/342 [00:00<?, ?B/s]
vocab.txt: 0%|          | 0.00/232k [00:00<?, ?B/s]
tokenizer.json: 0%|          | 0.00/712k [00:00<?, ?B/s]
special_tokens_map.json: 0%|          | 0.00/125 [00:00<?, ?B/s]
config.json: 0%|          | 0.00/191 [00:00<?, ?B/s]
```

```
[38]: #Embedding for the first chunk
embedding_1 = embedding_model.embed_query(document_chunks[0].page_content)
#Embedding for the second chunk
embedding_2 = embedding_model.embed_query(document_chunks[1].page_content)
```

```
[39]: print("Dimension of the embedding vector ",len(embedding_1))
len(embedding_1)==len(embedding_2)
```

Dimension of the embedding vector 1024

[39]: True

- The embedding model has provided a fixed length vector of 1024

0.5.5 Vector Database

```
[40]: # Create the output directory called med_db for storing the data
out_dir = 'med_db'

if not os.path.exists(out_dir):
    os.makedirs(out_dir)
```

```
[41]: # Creating the chroma vector store from the document chunks
vectorstore = Chroma.from_documents(
    document_chunks,
    embedding_model,
    persist_directory=out_dir
)
```

[42]: *# Loading the chroma vector store*

```
vectorstore =  
    ↪ Chroma(persist_directory=out_dir, embedding_function=embedding_model)
```

<ipython-input-42-7471167501cd>:2: LangChainDeprecationWarning: The class `Chroma` was deprecated in LangChain 0.2.9 and will be removed in 1.0. An updated version of the class exists in the :class:`~langchain-chroma` package and should be used instead. To use it run `pip install -U :class:`~langchain-chroma` and import as `from :class:`~langchain_chroma import Chroma``.

```
vectorstore =  
Chroma(persist_directory=out_dir, embedding_function=embedding_model)
```

[43]: *# Invoking the embedding function*

```
vectorstore.embeddings
```

[43]: HuggingFaceEmbeddings(client=SentenceTransformer(
 (0): Transformer({'max_seq_length': 512, 'do_lower_case': False}) with
 Transformer model: BertModel
 (1): Pooling({'word_embedding_dimension': 1024, 'pooling_mode_cls_token':
False, 'pooling_mode_mean_tokens': True, 'pooling_mode_max_tokens': False,
'pooling_mode_mean_sqrt_len_tokens': False, 'pooling_mode_weightedmean_tokens':
False, 'pooling_mode_lasttoken': False, 'include_prompt': True})
 (2): Normalize()
, model_name='thenlper/gte-large', cache_folder=None, model_kwargs={},
encode_kwargs={}, multi_process=False, show_progress=False)

[44]: *# Similarity search to find the top 4 matching documents*

```
relevant_docs = vectorstore.similarity_search("Medial Epicondylitis",k=4)
```

```
# Iterate through the relevant documents and print the wrapped content of each  
for doc in relevant_docs:  
    print(textwrap.fill(doc.page_content, width=120))  
    print("-" * 50) # Add a separator between documents
```

scar and degenerative tissue from the involved extensor tendons at the elbow. Surgery is usually considered only after at least 9 to 12 mo of unsuccessful conservative treatment. Medial Epicondylitis (Golfer's Elbow) Medial epicondylitis is inflammation of the flexor pronator muscle mass originating at the medial epicondyle of the elbow. Diagnosis is with provocative testing. Treatment is rest and ice and then exercises and gradual return to activity. Medial epicondylitis is caused by any activity that places a valgus force on the elbow or that involves forcefully flexing the volar forearm muscles, as occurs during pitching, golfing with improper technique, serving a tennis ball (particularly with top spin, with a racket that is too heavy or too tightly strung or has an undersized grip, or with heavy balls), and throwing a

javelin. Nonathletic activities that may cause The Merck Manual of Diagnosis & Therapy, 19th Edition Chapter 338.

Exercise & Sports Injury 3495 manandkn2016@gmail.com ZGIQC09SUD This file is meant for personal use by manandkn2016@gmail.com only. Sharing or publishing the contents in part or full is liable for legal action.

medial epicondylitis include bricklaying, hammering, and typing. Symptoms and Signs Pain occurs in the flexor pronator tendons (attached to the medial epicondyle) and in the medial epicondyle when the wrist is flexed or pronated against resistance. Diagnosis • Provocative testing To confirm the diagnosis, the examiner has the patient sit in a chair with the forearm resting on a table and the hand supinated. The patient tries to raise the fist by bending the wrist while the examiner holds it down. Pain around the medial epicondyle and in the flexor tendon origin confirms the diagnosis.

Treatment • Rest, ice, and muscle stretches • Modification of activity • Later, resistive exercises Treatment is symptomatic and similar to that of lateral epicondylitis (see p. 3300). Patients should avoid any activity that causes pain. Initially, rest, ice, NSAIDs, and stretching are used, occasionally with a corticosteroid injection into the painful area around the tendon. When pain subsides, gentle resistive exercises of the extensor and flexor muscles of the forearm are done, followed by eccentric and concentric resistive exercises. In general, surgery is considered only after at least 9 to 12 mo of failed conservative management. Surgical techniques to treat medial epicondylitis involve removing scar tissue and reattaching damaged tissues. Sidebar 338-3 Exercises to Strengthen the Wrist Flexors • Sit on a chair next to a table. • Place the forearm on the table, palm facing up, with the wrist and hand hanging over the edge.

- Hold a light weight in the hand.
- Slowly raise and lower the hand by bending and straightening the wrist.
- The set should last about 90 to 120 sec for rehabilitation and about 50 to 70 sec for general strength and conditioning.

Rest 1 min, then do additional sets until the forearms feel fatigued and worked. Repeat every 2 days while rehabilitating but only once every 7to10 days with normal or strong forearms and when using heavier weights and greater intensity of effort. If pain is felt, stop the exercise immediately, and try it again the next day.

diagnosis has been confirmed by MRI, surgical debridement or repair is the treatment of choice. Surgery is usually done arthroscopically. Lateral Epicondylitis (Tennis Elbow) Lateral epicondylitis results from inflammation and microtearing

of fibers in the extensor tendons of the forearm. Symptoms include pain at the lateral epicondyle of the elbow, which can radiate into the forearm. Diagnosis is by examination and provocative testing. Treatment is with rest, NSAIDs, and physical therapy. Theories about the pathophysiology of lateral epicondylitis include nonathletic and occupational activities that require repetitive and forceful forearm supination and pronation, as well as overuse or weakness (or both) of the extensor carpi radialis brevis and longus muscles of the forearm, which originate from the lateral epicondyle of the elbow. For example, during a backhand return in racket sports such as tennis, the elbow and wrist are extended, and the extensor tendons, particularly the extensor carpi radialis brevis, can be damaged when they roll over the lateral epicondyle and radial head. Contributing factors include weak shoulder and wrist muscles, a racket strung too tightly, an undersized grip, hitting heavy wet balls, and hitting off-center on the racket. In resistance trainees, injuries often are caused by overuse (too much activity or doing the same movements too often) or by muscle imbalance between the forearm extensors and flexors. Nonathletic activities that can cause or contribute to lateral epicondylitis include those involving grasping and twisting the elbow (eg, turning a screwdriver). With time, subperiosteal hemorrhage, calcification, spur formation on the lateral epicondyle, and, most importantly, tendon degeneration can occur. Symptoms and Signs Pain initially occurs in the extensor tendons of the forearm and around the lateral elbow when the wrist is extended against resistance (eg, as in using a manual screw driver or hitting a backhand shot with a racket). In resistance trainees, lateral epicondylitis is most noticeable during various rowing and chin-up exercises for the back muscles, particularly when the hands are pronated. Pain can extend from the

racket). In resistance trainees, lateral epicondylitis is most noticeable during various rowing and chin-up exercises for the back muscles, particularly when the hands are pronated. Pain can extend from the lateral epicondyle to the mid forearm. Diagnosis • Provocative testing Pain along the common extensor tendon when the fingers are extended against resistance and the elbow is held straight is diagnostic. Alternatively, the diagnosis is confirmed if the same pain occurs during the following maneuver: The patient sits on a chair with the forearm on the examination table and the elbow held flexed (bent) and the hand held palm downward; the examiner places a hand firmly on top of that of the patient, who tries to raise the hand by extending the wrist. Treatment • Rest,

ice, NSAIDs, extensor muscle stretches •
 Modification of activity • Later, resistive exercises Treatment involves a
 2-phased approach. Initially, rest, ice,
 NSAIDs, and stretching of the extensor muscles are used. Occasionally a
 corticosteroid injection into the painful area
 around the tendon is needed. When the pain subsides, gentle resistive exercises
 of the extensor (see Sidebar 338-2) and
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0.5.6 Retriever

```
[45]: # Retriever
retriever = vectorstore.as_retriever(
    search_type='similarity',
    search_kwargs={'k': 2}
)

[46]: rel_docs = retriever.get_relevant_documents("What are the symptoms of medial_
↳epicondylitis?")
print(rel_docs)
```

[Document(metadata={'producer': 'pdf-lib (https://github.com/Hopding/pdf-lib)',
 'format': 'PDF 1.7', 'total_pages': 4114, 'file_path': '/content/drive/MyDrive/C
 olab_Notebooks/Medical_Assistant_Project/medical_diagnosis_manual.pdf',
 'moddate': '2025-04-28T14:21:13+00:00', 'source': '/content/drive/MyDrive/Colab_
 Notebooks/Medical_Assistant_Project/medical_diagnosis_manual.pdf',
 'creationDate': 'D:20120615054440Z', 'creator': 'Atop CHM to PDF Converter',
 'trapped': '', 'subject': '', 'creationdate': '2012-06-15T05:44:40+00:00',
 'keywords': '', 'title': 'The Merck Manual of Diagnosis & Therapy, 19th
 Edition', 'page': 3504, 'author': '', 'modDate': 'D:20250428142113Z'},
 page_content="scar and degenerative tissue from the involved extensor tendons at
 the elbow. Surgery is usually\nconsidered only after at least 9 to 12 mo of
 unsuccessful conservative treatment.\nMedial Epicondylitis\n(Golfer's
 Elbow)\nMedial epicondylitis is inflammation of the flexor pronator muscle mass
 originating at the\nmedial epicondyle of the elbow. Diagnosis is with
 provocative testing. Treatment is rest and ice\nand then exercises and gradual
 return to activity.\nMedial epicondylitis is caused by any activity that places
 a valgus force on the elbow or that involves\nforcefully flexing the volar
 forearm muscles, as occurs during pitching, golfing with improper
 technique,\nserving a tennis ball (particularly with top spin, with a racket
 that is too heavy or too tightly strung or has\nan undersized grip, or with
 heavy balls), and throwing a javelin. Nonathletic activities that may cause\nThe
 Merck Manual of Diagnosis & Therapy, 19th Edition\nChapter 338. Exercise &

Sports Injury\n3495\nmanandkn2016@gmail.com\nZGIQC09SUD\nThis file is meant for personal use by manandkn2016@gmail.com only.\nSharing or publishing the contents in part or full is liable for legal action."), Document(metadata={'producer': 'pdf-lib (https://github.com/Hopding/pdf-lib)', 'author': '', 'title': 'The Merck Manual of Diagnosis & Therapy, 19th Edition', 'file_path': '/content/drive/MyDrive/Colab_Notebooks/Medical_Assistant_Project/medical_diagnosis_manual.pdf', 'page': 3505, 'total_pages': 4114, 'moddate': '2025-04-28T14:21:13+00:00', 'subject': '', 'source': '/content/drive/MyDrive/Colab_Notebooks/Medical_Assistant_Project/medical_diagnosis_manual.pdf', 'modDate': 'D:20250428142113Z', 'creator': 'Atop CHM to PDF Converter', 'trapped': '', 'format': 'PDF 1.7', 'creationdate': '2012-06-15T05:44:40+00:00', 'keywords': '', 'creationDate': 'D:20120615054440Z'}, page_content='medial epicondylitis include bricklaying, hammering, and typing.\nSymptoms and Signs\nPain occurs in the flexor pronator tendons (attached to the medial epicondyle) and in the medial\nepicondyle when the wrist is flexed or pronated against resistance.\nDiagnosis\n• Provocative testing\nTo confirm the diagnosis, the examiner has the patient sit in a chair with the forearm resting on a table\nand the hand supinated. The patient tries to raise the fist by bending the wrist while the examiner holds it\ndown. Pain around the medial epicondyle and in the flexor tendon origin confirms the diagnosis.\nTreatment\n• Rest, ice, and muscle stretches\n• Modification of activity\n• Later, resistive exercises\nTreatment is symptomatic and similar to that of lateral epicondylitis (see p. 3300). Patients should avoid\nany activity that causes pain. Initially, rest, ice, NSAIDs, and stretching are used, occasionally with a\ncorticosteroid injection into the painful area around the tendon. When pain subsides, gentle resistive\nexercises of the extensor and flexor muscles of the forearm are done, followed by eccentric and\nconcentric resistive exercises. In general, surgery is considered only after at least 9 to 12 mo of failed\nconservative management. Surgical techniques to treat medial epicondylitis involve removing scar tissue\nand reattaching damaged tissues.\nSidebar 338-3 Exercises to Strengthen the Wrist Flexors\n• Sit on a chair next to a table.\n• Place the forearm on the table, palm facing up, with the wrist and hand hanging over the edge.\n• Hold a light weight in the hand.\n• Slowly raise and lower the hand by bending and straightening the wrist.\n• The set should last about 90 to 120 sec for rehabilitation and about 50 to 70 sec for general strength\nand conditioning. Rest 1 min, then do additional sets until the forearms feel fatigued and worked.\nRepeat every 2 days while rehabilitating but only once every 7to10 days with normal or strong forearms\nand when using heavier weights and greater intensity of effort. If pain is felt, stop the exercise\nimmediately, and try it again the next day.'])]

```
<ipython-input-46-2e050441f599>:1: LangChainDeprecationWarning: The method
`BaseRetriever.get_relevant_documents` was deprecated in langchain-core 0.1.46
and will be removed in 1.0. Use `meth:~invoke` instead.
```

```
rel_docs = retriever.get_relevant_documents("What are the symptoms of medial
epicondylitis?")
```

```
[47]: model_output = llm(
    "What are the symptoms of medial epicondylitis?",
```

```

    max_tokens=512,
    temperature=0,
)

```

Llama.generate: prefix-match hit

```
[48]: print(textwrap.fill(model_output['choices'][0]['text'], width=120))
```

Medial epicondylitis, also known as golfer's elbow, is a condition characterized by pain and inflammation in the forearm, specifically at the medial epicondyle, which is the bony prominence on the inside of the elbow. The symptoms of medial epicondylitis include: 1. Pain and tenderness on the inner side of the elbow, especially with gripping or flexing the wrist. 2. Weakness in the hand and wrist, making it difficult to perform tasks that require a strong grip. 3. Pain radiating down the forearm towards the wrist. 4. Swelling or bruising in the affected area. 5. Limited range of motion in the elbow and wrist. 6. Aggravation of symptoms with activities such as lifting, carrying heavy objects, or playing sports that require repetitive motions. The pain and discomfort caused by medial epicondylitis can make it difficult to perform daily activities, and in severe cases, may require medical intervention. If you are experiencing any of these symptoms, it is important to consult a healthcare professional for an accurate diagnosis and treatment plan.

0.5.7 System and User Prompt Template

```
[49]: qna_system_message = """
You are an helpful AI medical assistant specialized in providing information_
↳from the 19th edition of Merck Manual of Medical Diagnosis.

Answer the question based only on the following context from the Merck Manual_
↳of Medical Diagnosis.

Do not make up information or provide answers from outside the given context.

User input will have the context required by you to answer user questions.

Be concise and accurate. If listing steps, drugs or symptoms - present them_
↳with clarity.

This context will begin with the token: ###Context.

The context contains references to specific portions of a document relevant to_
↳the user query.
```

User questions will begin with the token: ###Question.

Please answer only using the context provided in the input. Do not mention
↳ anything about the context or the Merck Manual of Medical Diagnosis in your
↳ final answer.

If the answer is not found in the context, respond "I don't know".
"""

```
[50]: qna_user_message_template = """
###Context
Following are the relevant information from the reference:
{context}

###Question
{question}
"""
```

0.5.8 Response Function

```
[51]: def generate_rag_response(user_input,k=3,max_tokens=512,temperature=0,top_p=0.
↳95,top_k=50):
    global qna_system_message,qna_user_message_template
    # Retrieve relevant document chunks
    relevant_document_chunks = retriever.
    ↳get_relevant_documents(query=user_input,k=k)
    context_list = [d.page_content for d in relevant_document_chunks]

    # Combine document chunks into a single context
    context_for_query = ". ".join(context_list)

    user_message = qna_user_message_template.replace('{context}',
    ↳context_for_query)
    user_message = user_message.replace('{question}', user_input)

    prompt = qna_system_message + '\n' + user_message

    # Generate the response
    try:
        response = llm(
            prompt=prompt,
            max_tokens=max_tokens,
            temperature=temperature,
            top_p=top_p,
            top_k=top_k
        )
```

```

    # Extract and print the model's response
    response = response['choices'][0]['text'].strip()
except Exception as e:
    response = f'Sorry, I encountered the following error: \n {e}'

return response

```

0.6 Question Answering using RAG

Query 1: What is the protocol for managing sepsis in a critical care unit?

```

[52]: start_time = datetime.now()

response_string_rag_1 = textwrap.fill(generate_rag_response(user_query_1),
    ↪width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_1)

```

Llama.generate: prefix-match hit

Time taken for query response: 23.398886 seconds

The context provides information on the management of septic shock in a critical care unit. The following steps should be taken: 1. Monitor vital signs, fluid intake and output, blood glucose levels, lactate and electrolyte levels, renal function, and possibly sublingual PCO₂ frequently. 2. Administer normal saline for fluid resuscitation until CVP reaches 8 mm Hg (10 cm H₂O) or PAOP reaches 12 to 15 mm Hg. Oliguria with hypotension is not a contraindication to vigorous fluid resuscitation. 3. If the patient remains hypotensive after CVP or PAOP has been raised to target levels, dopamine may be given to increase mean BP to at least 60 mm Hg. If dopamine dose exceeds 20 g/kg/min, another vasopressor, typically norepinephrine, may be added. 4. Administer oxygen by mask or nasal prongs and consider tracheal intubation and mechanical ventilation for respiratory failure. 5. Normalize blood glucose levels to improve outcome in critically ill patients using a continuous IV insulin infusion titrated to maintain glucose between 80 to 110 mg/dL (4.4 to 6.1 mmol/L). 6. Administer replacement-dose corticosteroids. 7. Obtain blood, body fluids, and wound sites for Gram stain and culture before administering parenteral antibiotics. 8

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
[53]: start_time = datetime.now()

response_string_rag_2 = textwrap.fill(generate_rag_response(user_query_2),
    ↪width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_2)
```

Llama.generate: prefix-match hit

Time taken for query response: 7.342931 seconds

The context mentions that appendicitis is characterized by abdominal pain, anorexia, and abdominal tenderness. The treatment for appendicitis is surgical removal of the appendix. There is no mention of any curative medicine for appendicitis in the context.

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
[54]: start_time = datetime.now()

response_string_rag_3 = textwrap.fill(generate_rag_response(user_query_3),
    ↪width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_3)
```

Llama.generate: prefix-match hit

Time taken for query response: 15.043351 seconds

Based on the context provided, alopecia areata is a condition characterized by sudden patchy hair loss. The scalp and beard are most frequently affected areas, but any hairy area may be involved. It is thought to be an autoimmune disorder affecting genetically susceptible people exposed to unclear environmental triggers. Treatment options for alopecia areata include topical, intralesional, or systemic corticosteroids, topical minoxidil, topical anthralin, topical

immunotherapy (diphencyprone or squaric acid dibutylester), or psoralen plus ultraviolet A (PUVA). The efficacy is usually evident within 6 to 8 months of treatment. Common practice is to continue treatment for as long as positive results persist, and once treatment is stopped, hair loss returns to previous levels. Other causes of sudden patchy hair loss are treated by addressing the underlying disorder.

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
[55]: start_time = datetime.now()

response_string_rag_4 = textwrap.fill(generate_rag_response(user_query_4),
    width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_4)
```

Llama.generate: prefix-match hit

Time taken for query response: 18.584878 seconds

Based on the context provided, the following treatments are recommended for a person who has sustained a physical injury

to brain tissue: 1. Ensuring a reliable airway and maintaining adequate ventilation, oxygenation, and blood pressure.

2. Surgery to place monitors to track and treat intracranial pressure, decompress the brain if intracranial pressure is increased, or remove intracranial hematomas. 3. Maintaining adequate brain perfusion and oxygenation and preventing complications of altered sensorium in the first few days after the injury. 4.

Rehabilitation to help recover functional abilities and prevent secondary disabilities such as pressure ulcers, joint contractures, pneumonia, etc. 5. Cognitive therapy for patients with severe cognitive dysfunction, which is often begun immediately after the injury and continued

for months or years. 6. Prevention of complications during the acute phase, including measures to prevent contractures,

pressure ulcers, and pneumonia. 7. A team approach that combines physical, occupational, and speech therapy, skill-

building activities, and counseling to meet the patient's social and emotional needs. 8. Family education and support

from brain injury support groups. 9. For patients whose coma exceeds 24 hours, a prolonged period of rehabilitation,

particularly in cognitive and emotional areas, may be required.

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
[56]: start_time = datetime.now()

response_string_rag_5 = textwrap.fill(generate_rag_response(user_query_5),
    ↪width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_5)
```

Llama.generate: prefix-match hit

Time taken for query response: 17.91738 seconds

Based on the context provided, here is the answer: The person with a suspected fractured leg requires sterile wound dressings, tetanus prophylaxis, and broad-spectrum antibiotics. They should seek medical care as soon as possible if an odor emanates from within the cast or if a fever develops, which may indicate infection. Good hygiene is important. A splint can be used to immobilize some stable injuries, including suspected but unproven fractures. Patients should apply ice and move more with a splint, as it does not contribute to compartment syndrome. Prolonged immobilization of a joint can cause stiffness, contractures, and muscle atrophy, particularly in the elderly. Some rapidly healing injuries are best treated with resumption of active motion within the first few days or weeks (early mobilization). Fractures are cracks in bones that cause symptoms such as pain, swelling, ecchymosis, crepitation, deformity, and abnormal motion. Occasional complications include fat embolism, arterial injury, compartment syndrome, nerve injuries, and infection. Diagnosis is by clinical criteria and usually plain x-rays. Treatment involves analgesics, immobilization, and sometimes surgery. The person should use a cane when descending stairs with their bad leg, moving it to the lower step shortly before descending. They should also avoid injuring the skin under the cast and seek medical care if an odor or fever develops.

0.6.1 Observations on using RAG

- The answers provided for the five queries are clear, concise, and focused, without any unnecessary information.
- The answers are context specific and accurate.

0.6.2 Fine-tuning

Query 1: What is the protocol for managing sepsis in a critical care unit?

```
[57]: start_time = datetime.now()

response_string_rag_ft_1 = textwrap.fill(generate_rag_response(user_query_1,
    ↳temperature = 0.5), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_ft_1)
```

Llama.generate: prefix-match hit

Time taken for query response: 20.715468 seconds

Answer: Patients with sepsis should be treated in an ICU and monitored frequently for systemic pressure, CVP or PAOP, pulse oximetry, ABGs, blood glucose, lactate, electrolyte levels, renal function, sublingual PCO₂, urine output, and oxygen saturation. Fluid resuscitation with 0.9% saline should be given until CVP reaches 8 mm Hg (10 cm H₂O) or PAOP reaches 12 to 15 mm Hg. If a patient remains hypotensive after fluid resuscitation, dopamine may be given to increase mean BP to at least 60 mm Hg. Oxygen should be provided via mask or nasal prongs, and tracheal intubation and mechanical ventilation may be needed for respiratory failure. Normalization of blood glucose levels improves outcome in critically ill patients and can be maintained with a continuous IV insulin infusion titrated to keep glucose between 80 to 110 mg/dL (4.4 to 6.1 mmol/L). Antibiotics should be given promptly based on suspected source, clinical setting, and previous culture results, and continued for at least 5 days after shock resolves. Septic foci must also be eliminated through drainage or surgical excision of necrotic tissues.

```
[58]: #Change the temperature to 1.5 and rerun the prompt
start_time = datetime.now()

response_string_rag_ft_1 = textwrap.fill(generate_rag_response(user_query_1,
    ↳temperature = 1.5), width=120)
```

```

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_ft_1)

```

Llama.generate: prefix-match hit

Time taken for query response: 18.151912 seconds

You should maintain the patient's systemic pressure, CVP, PAOP or both, pulse oximetry, ABGs, blood glucose, lactate and electrolyte levels, renal function, and sublingual PCO2 under frequent monitoring. Administer fluid resuscitation with 0.9% saline until the CVP reaches 8 mm Hg (10 cm H2O) or PAOP reaches 12 to 15 mm Hg. Use dopamine to increase mean BP if needed, and consider adding another vasopressor if the dose exceeds 20 µg/kg/min. Administer O2 by mask or nasal prongs and perform tracheal intubation and mechanical ventilation for respiratory failure as needed. Use gentle and frequent blood tests to detect problems early, including daily electrolytes and a CBC, Mg, phosphate, Ca levels for arrhythmias, and weekly liver enzymes and coagulation profiles for patients on TPN. Implement prompt empiric therapy with antibiotics when sepsis is suspected. A recommended regimen includes gentamicin or tobramycin with a 3rd-generation cephalosporin or ceftazidime with a fluoroquinolone if Pseudomonas is suspected, or vancomycin if resistant staphylococci or enterococci are suspected. Normalize the patient's blood glucose levels to improve outcome. A continuous IV insulin infusion is used to maintain blood glucose between 80 to 110 mg/d

- The **temperature** parameter significantly influences the LLM's response to the query.
- temperature = 0.5 produces a more factual, concise response (more safer)
- temperature = 1.5 produces a more verbose, creative response (more risk of inaccuracies creeping in)
- For a medical protocol, lower temperatures (even 0) could be preferred to provide factual, accurate responses that promote adherence to established knowledge.

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```

[59]: start_time = datetime.now()

response_string_rag_ft_2 = textwrap.fill(generate_rag_response(user_query_2,
↪top_p = 0.95), width=120)

```

```

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_ft_2)

```

Llama.generate: prefix-match hit

Time taken for query response: 7.005253 seconds

The context mentions that appendicitis is characterized by abdominal pain, anorexia, and abdominal tenderness. The treatment for appendicitis is surgical removal of the appendix. Antibiotics can improve survival rate if surgery is impossible.

[76]: *# Change the top_p = 2 and rerun the prompt*

```

start_time = datetime.now()

response_string_rag_ft_2 = textwrap.fill(generate_rag_response(user_query_2,
    ↪top_p = 2.0), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_ft_2)

```

Llama.generate: prefix-match hit

Time taken for query response: 10.408903 seconds

The context mentions that appendicitis is characterized by abdominal pain, anorexia, and abdominal tenderness. It also states that treatment for appendicitis is surgical removal. Therefore, the answer would be: Appendicitis presents with symptoms such as abdominal pain, loss of appetite (anorexia), and abdominal tenderness. There is no cure for appendicitis through medication alone, and the standard treatment is surgical removal via open or laparoscopic appendectomy.

- A **top_p** value is typically between 0 and 1. 0 is excluded and 1 is included.
- The value used here for top_p (2) is outside the range and probably replaced with 1. This could be the reason that the responses are very similar.
- The second response is slightly more verbose.

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the

possible causes behind it?

```
[61]: start_time = datetime.now()

response_string_rag_ft_3 = textwrap.fill(generate_rag_response(user_query_3,
↳top_k = 50), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_ft_3)
```

Llama.generate: prefix-match hit

Time taken for query response: 22.870857 seconds

Based on the context, alopecia areata is a common cause of sudden patchy hair loss. The treatment options for alopecia areata include topical, intralesional, or systemic corticosteroids, topical minoxidil, topical anthralin, topical immunotherapy (diphencyprone or squaric acid dibutylester), or psoralen plus ultraviolet A (PUVA). The efficacy of these treatments is usually evident within 6 to 8 months. Adverse effects include decreased libido, erectile and ejaculatory dysfunction, hypersensitivity reactions, gynecomastia, and myopathy for systemic corticosteroids and minoxidil. There may be a decrease in prostate-specific antigen levels in older men taking minoxidil, which should be taken into account when that test is used for cancer screening. Common practice is to continue treatment for as long as positive results persist. Once treatment is stopped, hair loss returns to previous levels. Finasteride is not indicated for women and is contraindicated in pregnant women because it has teratogenic effects in animals. Hormonal modulators such as oral contraceptives or spironolactone may be useful for female-pattern hair loss associated with hyperandrogenemia. Surgical options include follicle transplant, scalp flaps, and alopecia reduction. Few procedures have been subjected to scientific scrutiny, but patients who are self-conscious about their hair loss may consider them. Other causes of sudden patchy hair loss should be treated based on the underlying disorder.

```
[62]: # Change top_k to 35 and rerun the prompt
start_time = datetime.now()

response_string_rag_ft_3 = textwrap.fill(generate_rag_response(user_query_3,
↳top_k = 35), width=120)
```

```

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_ft_3)

```

Llama.generate: prefix-match hit

Time taken for query response: 17.802157 seconds

Based on the context provided, alopecia areata is a condition characterized by sudden patchy hair loss. The scalp and beard are most frequently affected areas. Alopecia areata is thought to be an autoimmune disorder affecting genetically susceptible people exposed to unclear environmental triggers. Treatment options for alopecia areata include topical, intralesional, or systemic corticosteroids, topical minoxidil, topical anthralin, topical immunotherapy (diphencyprone or squaric acid dibutylester), or psoralen plus ultraviolet A (PUVA). Efficacy is usually evident within 6 to 8 months of treatment. Adverse effects include decreased libido, erectile and ejaculatory dysfunction, hypersensitivity reactions, gynecomastia, and myopathy. Common practice is to continue treatment for as long as positive results persist. Once treatment is stopped, hair loss returns to previous levels. Finasteride is not indicated for women and is contraindicated in pregnant women because it has teratogenic effects in animals. Hormonal modulators such as oral contraceptives or spironolactone may be useful for female-pattern hair loss associated with hyperandrogenemia. Surgical options include follicle transplant, scalp flaps, and alopecia reduction. Few procedures have been subjected to scientific scrutiny, but patients who are self-conscious about their hair loss may consider them. Other causes of sudden patchy hair loss should be evaluated and treated accordingly.

- **top_k** sampling is expected to restrict the model's choice for the next k probable tokens. Smaller k is supposed to make the output more focussed and less random.
- In our case here, we used k = 50 and then k = 35. Though they are large numbers, the responses are very similar as k = 35 will steer the answer towards most probable sequence. The model is confident of its standard answer.
- Maybe we need to use more restrictive values closer to 1 or 2.

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

[63]: start_time = datetime.now()

```

response_string_rag_ft_4 = textwrap.fill(generate_rag_response(user_query_4,
↳max_tokens = 512), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_ft_4)

```

Llama.generate: prefix-match hit

Time taken for query response: 16.702748 seconds

Answer: The initial treatment consists of ensuring a reliable airway and maintaining adequate ventilation, oxygenation, and blood pressure. Surgery may be needed to place monitors to track and treat intracranial pressure, decompress the brain if intracranial pressure is increased, or remove intracranial hematomas. In the first few days after the injury, maintaining adequate brain perfusion and oxygenation and preventing complications of altered sensorium are important. Subsequently, many patients require rehabilitation to regain function. Rehabilitation specialists evaluate patients early to establish baseline findings and prioritize treatment based on cognitive dysfunction, muscle weakness, spasticity, incoordination, and ataxia. Prevention of secondary disabilities, such as pressure ulcers and joint contractures, is important during the acute phase. Daily routine care includes measures to prevent contractures, pressure ulcers, and pneumonia. A team approach that combines physical, occupational, and speech therapy, skill-building activities, and counseling is used to meet the patient's social and emotional needs. For patients whose coma exceeds 24 hours, a prolonged period of rehabilitation may be required, particularly in cognitive and emotional areas.

```

[64]: # Change the max_tokens to 1024 and rerun the prompt
start_time = datetime.now()

response_string_rag_ft_4 = textwrap.fill(generate_rag_response(user_query_4,
↳max_tokens = 1024), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_ft_4)

```

Llama.generate: prefix-match hit

Time taken for query response: 9.791712 seconds

Based on the context provided, the following treatments are recommended for a person who has sustained a physical injury

to brain tissue: 1. Ensuring a reliable airway and maintaining adequate ventilation, oxygenation, and blood pressure.

2. Surgery to place monitors to track and treat intracranial pressure, decompress the brain if intracranial pressure is

increased, or remove intracranial hematomas. 3. Maintaining adequate brain perfusion and oxygenation and preventing

complications of altered sensorium in the first few days after the injury. 4.

Rehabilitation services, including

prevention of secondary disabilities, prevention of pneumonia, family education, cognitive therapy, and a team approach

that combines physical, occupational, and speech therapy. 5. For patients whose coma exceeds 24 hours, a prolonged

period of rehabilitation, particularly in cognitive and emotional areas, is often required.

- With `max_length = 512`, the LLM provided a useful but condensed set of instructions in the response.
- With `max_length = 1024`, the LLM leveraged a larger token limit and provided a slightly more detailed recovery/treatment steps.
- Both responses were quite accurate advice for managing a TBI.

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
[65]: start_time = datetime.now()

response_string_rag_ft_5 = textwrap.fill(generate_rag_response(user_query_5, k=
↳ 3), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_ft_5)
```

Llama.generate: prefix-match hit

Time taken for query response: 10.762808 seconds

Based on the context provided, here is the answer: The person with a suspected fractured leg requires sterile wound

dressings, tetanus prophylaxis, and broad-spectrum antibiotics. They should use a splint to immobilize the injury if

it's stable and allow for active motion, ice application, and mobility without

contributing to compartment syndrome. It's essential to maintain good hygiene and seek medical care promptly if there's an odor from within the cast or a fever develops, which may indicate infection. Prolonged immobilization can lead to complications such as stiffness, contractures, and muscle atrophy, so early mobilization is recommended for rapidly healing injuries. The Merck Manual suggests consulting specific sections of the publication for more detailed information on fractures of various body parts.

```
[66]: # Change k = 5 and rerun the prompt
start_time = datetime.now()

response_string_rag_ft_5 = textwrap.fill(generate_rag_response(user_query_5, k=
    ↪ 5), width=120)

end_time = datetime.now()

elapsed_time = end_time - start_time

print(f"Time taken for query response: {elapsed_time.total_seconds()} seconds")
print(response_string_rag_ft_5)
```

Llama.generate: prefix-match hit

Time taken for query response: 17.827406 seconds

Based on the context provided, here's the answer: 1. The person with a suspected fractured leg requires sterile wound dressings, tetanus prophylaxis, and broad-spectrum antibiotics (eg, a 2nd-generation cephalosporin plus an aminoglycoside). 2. They should be moved to lower steps with the help of a cane or crutches while descending. 3. The leg should be immobilized using a splint to allow for application of ice and movement, preventing further injury and complications like compartment syndrome. 4. Prolonged immobilization (more than 3 to 4 weeks) may cause stiffness, contractures, and muscle atrophy, especially in the elderly. Early mobilization is recommended for rapidly healing injuries to minimize these complications. 5. The person should be advised to keep the splint dry, never put objects inside it, inspect the edges and skin around the splint daily, and apply lotion to any red or sore areas. 6. If severe swelling is likely, the cast (and all padding) may need to be cut open from end to end medially and laterally (bivalved). 7. The person should elevate their injured leg above heart level for the first 2 days to minimize swelling and relieve

- Increasing the number of retrieved documents (k) from 3 to 5 for the RAG system had a clear positive impact on the comprehensiveness and detail of the answer for Query 5.
- When k=3, the system provided a good, relevant summary based on a more limited set of information.
- When k=5, the system was able to draw from a richer contextual base, resulting in a more detailed, specific, and actionable answer that included more nuances of fracture management as likely described in the Merck Manual.
- If the additional retrieved chunks are relevant, the LLM has more raw material to work with, leading to better responses. However, there's a balance; retrieving too many chunks, especially if some are irrelevant, can introduce noise or exceed the LLM's context window capacity. In this case, moving from 3 to 5 appears beneficial

0.7 Output Evaluation

```
[67]: groundedness_rater_system_message = """
You are tasked with rating AI generated answers to questions posed by users.
You will be presented a question, context used by the AI system to generate the
    ↪answer and an AI generated answer to the question.
In the input, the question will begin with ###Question, the context will begin
    ↪with ###Context while the AI generated answer will begin with ###Answer.

Evaluation criteria:
The task is to judge the extent to which the metric is followed by the answer.
1 - The metric is not followed at all
2 - The metric is followed only to a limited extent
3 - The metric is followed to a good extent
4 - The metric is followed mostly
5 - The metric is followed completely

Metric:
The answer should be derived only from the information presented in the context

Instructions:
1. First write down the steps that are needed to evaluate the answer as per the
    ↪metric.
2. Give a step-by-step explanation if the answer adheres to the metric
    ↪considering the question and context as the input.
3. Next, evaluate the extent to which the metric is followed.
4. Use the previous information to rate the answer using the evaluation criteria
    ↪and assign a score.
"""
```

```
[68]: relevance_rater_system_message = """
You are tasked with rating AI generated answers to questions posed by users.
You will be presented a question, context used by the AI system to generate the
    ↪answer and an AI generated answer to the question.
```

In the input, the question will begin with ###Question, the context will begin with ###Context while the AI generated answer will begin with ###Answer.

Evaluation criteria:

The task is to judge the extent to which the metric is followed by the answer.

- 1 - The metric is not followed at all
- 2 - The metric is followed only to a limited extent
- 3 - The metric is followed to a good extent
- 4 - The metric is followed mostly
- 5 - The metric is followed completely

Metric:

Relevance measures how well the answer addresses the main aspects of the question, based on the context.

Consider whether all and only the important aspects are contained in the answer when evaluating relevance.

Instructions:

- 1. First write down the steps that are needed to evaluate the context as per the metric.
- 2. Give a step-by-step explanation if the context adheres to the metric considering the question as the input.
- 3. Next, evaluate the extent to which the metric is followed.
- 4. Use the previous information to rate the context using the evaluation criteria and assign a score.

"""

```
[69]: user_message_template = """
```

```
###Question
{question}
```

```
###Context
{context}
```

```
###Answer
{answer}
"""
```

```
[70]: def
    generate_ground_relevance_response(user_input,k=2,max_tokens=512,temperature=0,top_p=0.
    95,top_k=50):
    global qna_system_message,qna_user_message_template
    # Retrieve relevant document chunks
    relevant_document_chunks = retriever.
    get_relevant_documents(query=user_input,k=k)
    context_list = [d.page_content for d in relevant_document_chunks]
```

```

context_for_query = ". ".join(context_list)

# Combine user_prompt and system_message to create the prompt
prompt = f"""[INST]{qna_system_message}\n
{'user'}: {qna_user_message_template.
↪format(context=context_for_query, question=user_input)}
[/INST]"""

response = llm(
    prompt=prompt,
    max_tokens=max_tokens,
    temperature=temperature,
    top_p=top_p,
    top_k=top_k,
    stop=['INST'],
    echo=False
)

answer = response["choices"][0]["text"]

# Combine user_prompt and system_message to create the prompt
groundedness_prompt = f"""[INST]{groundedness_rater_system_message}\n
{'user'}: {user_message_template.
↪format(context=context_for_query, question=user_input, answer=answer)}
[/INST]"""

# Combine user_prompt and system_message to create the prompt
relevance_prompt = f"""[INST]{relevance_rater_system_message}\n
{'user'}: {user_message_template.
↪format(context=context_for_query, question=user_input, answer=answer)}
[/INST]"""

response_1 = llm(
    prompt=groundedness_prompt,
    max_tokens=max_tokens,
    temperature=temperature,
    top_p=top_p,
    top_k=top_k,
    stop=['INST'],
    echo=False
)

response_2 = llm(
    prompt=relevance_prompt,
    max_tokens=max_tokens,
    temperature=temperature,
    top_p=top_p,

```

```

        top_k=top_k,
        stop=['INST'],
        echo=False
    )

    return response_1['choices'][0]['text'],response_2['choices'][0]['text']

```

Query 1: What is the protocol for managing sepsis in a critical care unit?

```

[71]: ground,rel = _
      ↳ generate_ground_relevance_response(user_input=user_query_1,max_tokens=512)

      #print(ground,end="\n\n")
      #print(rel)

      print(textwrap.fill(ground,width=120),end="\n\n")
      print(textwrap.fill(rel,width=120))

```

Llama.generate: prefix-match hit
 Llama.generate: prefix-match hit
 Llama.generate: prefix-match hit

Steps to evaluate the answer: 1. Identify the information in the context related to managing sepsis in a critical care unit. 2. Check if the AI generated answer includes all the steps identified in step 1. 3. Verify that the information in the answer is derived only from the context and not from any external sources. Explanation: The context provides detailed information about managing sepsis in a critical care unit, including monitoring parameters, fluid resuscitation, use of vasopressors, oxygen therapy, normalization of blood glucose levels, and replacement-dose corticosteroids. The AI generated answer includes all these steps, making it clear that the answer is derived directly from the context. Evaluation: The metric is followed completely as the answer is derived only from the information presented in the context. Rating: Based on the evaluation criteria, I would rate the answer a 5 for following the metric completely.

Steps to evaluate the context as per the relevance metric: 1. Identify the main aspects of the question: managing sepsis in a critical care unit. 2. Read through the context to understand the information provided about managing sepsis in a critical care unit. 3. Check if all important aspects of managing sepsis in a critical care unit are present in the context. 4. Determine if any irrelevant information is included in the context that does not relate to managing sepsis in a critical care unit. The context adheres to the relevance metric

considering the question as follows: 1. The context mentions treating patients with septic shock in an ICU and monitoring various physiologic parameters, which is directly related to managing sepsis in a critical care unit. 2. The context provides information on fluid resuscitation, use of vasopressors, oxygen therapy, normalization of blood glucose levels, and replacement-dose corticosteroids, all of which are important aspects of managing sepsis in a critical care unit. 3. There is no irrelevant information included in the context that does not relate to managing sepsis in a critical care unit. The metric is followed mostly as all important aspects of managing sepsis in a critical care unit are present in the context, and there is no irrelevant information included. Therefore, I would rate the context a 4 on the evaluation criteria scale.

Query 2: What are the common symptoms for appendicitis, and can it be cured via medicine? If not, what surgical procedure should be followed to treat it?

```
[72]: ground,rel = generate_ground_relevance_response(user_input=user_query_2,max_tokens=512)

#print(ground,end="\n\n")
#print(rel)

print(textwrap.fill(ground,width=120),end="\n\n")
print(textwrap.fill(rel,width=120))
```

```
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
```

Steps to evaluate the answer: 1. Identify the key information in the context related to appendicitis and its symptoms as well as its treatment. 2. Check if the AI generated answer includes only the information derived from the context. 3. Evaluate the extent to which the metric is followed. Explanation: The AI generated answer correctly identifies the common symptoms for appendicitis, which are mentioned in the context. It also states that the treatment for appendicitis is surgical removal of the appendix, which is also stated in the context. The AI generated answer does not include any additional information or incorrect statements. Therefore, the metric is followed completely. Rating: Based on the evaluation criteria and the complete adherence to the metric, I would rate the answer as 5.

Steps to evaluate the context as per the relevance metric: 1. Identify the main aspects of the question: common

symptoms for appendicitis and whether it can be cured via medicine or if surgery is required. 2. Determine if the context provides information on these main aspects. 3. Check if all important aspects are contained in the context and if any irrelevant information is present. The context adheres to the metric considering the question as follows: 1. The context mentions common symptoms for appendicitis, which includes abdominal pain, anorexia (loss of appetite), and abdominal tenderness. 2. It also states that appendicitis cannot be cured via medicine alone and requires surgical removal. 3. The context provides detailed information on the treatment options for appendicitis, including open or laparoscopic appendectomy. 4. No irrelevant information is present in the context. The metric is followed mostly as all important aspects are contained in the context and no irrelevant information is present. Rating: 4 (The metric is followed mostly)

Query 3: What are the effective treatments or solutions for addressing sudden patchy hair loss, commonly seen as localized bald spots on the scalp, and what could be the possible causes behind it?

```
[73]: ground,rel =  
    ↳generate_ground_relevance_response(user_input=user_query_3,max_tokens=512)  
  
#print(ground,end="\n\n")  
#print(rel)  
  
print(textwrap.fill(ground,width=120),end="\n\n")  
print(textwrap.fill(rel,width=120))
```

Llama.generate: prefix-match hit
Llama.generate: prefix-match hit
Llama.generate: prefix-match hit

Steps to evaluate the answer: 1. Identify the main question and the specific information being asked for in the question. 2. Read through the context provided to identify any relevant information related to the question. 3. Determine if the AI generated answer is derived only from the information presented in the context. Explanation: The main question asks about effective treatments or solutions for addressing sudden patchy hair loss and possible causes behind it. The context provides detailed information about various types of hair loss, their causes, and treatments. The AI generated answer correctly identifies that sudden patchy hair loss is known as alopecia areata and lists several treatment options for this condition, including topical corticosteroids, minoxidil, anthralin, immunotherapy, systemic

corticosteroids, oral antimalarials, retinoids, or immunosuppressants. The answer also mentions that the cause of alopecia areata is believed to be an autoimmune disorder affecting genetically susceptible individuals exposed to unclear environmental triggers. The information in the AI generated answer is directly derived from the context provided, as it references several treatments mentioned in the context and explains the cause of alopecia areata based on the information presented in the context. Evaluation: Based on the evaluation criteria, I would rate this answer as a 5 because the metric is followed completely. The AI generated answer is derived only from the information presented in the context and accurately reflects the information provided in the context.

Steps to evaluate the context as per the relevance metric: 1. Identify the main aspects of the question: effective treatments or solutions for addressing sudden patchy hair loss and possible causes. 2. Determine if the context provides information on these main aspects. 3. Check if all important aspects are contained in the context and if any irrelevant information is present. The context adheres to the metric considering the question as follows: 1. The context mentions various treatments for sudden patchy hair loss, including topical corticosteroids, minoxidil, anthralin, immunotherapy, oral antimalarials, corticosteroids, retinoids, and immunosuppressants. This directly addresses the question's first aspect (effective treatments or solutions). 2. The context also discusses possible causes of sudden patchy hair loss, such as autoimmune disorders, underlying disorders, and environmental triggers. This addresses the second aspect of the question. 3. No irrelevant information is present in the context. The metric is followed mostly since all important aspects are addressed, but some additional details could be provided to make the answer more comprehensive (e.g., success rates, side effects, or long-term management strategies). Based on the evaluation criteria: 1 - The metric is not followed at all: 1 2 - The metric is followed only to a limited extent: 2 3 - The metric is followed to a good extent: 3 4 - The metric is followed mostly: 4 5 - The metric is followed completely: 5 Given the information above, I would rate the context as a 4 (followed mostly) on the evaluation criteria.

Query 4: What treatments are recommended for a person who has sustained a physical injury to brain tissue, resulting in temporary or permanent impairment of brain function?

```
[74]: #print(ground,end="\n\n")
      #print(rel)
```

```
print(textwrap.fill(ground,width=120),end="\n\n")
print(textwrap.fill(rel,width=120))
```

Steps to evaluate the answer: 1. Identify the main question and the specific information being asked for in the question. 2. Read through the context provided to identify any relevant information related to the question. 3. Determine if the AI generated answer is derived only from the information presented in the context. Explanation: The main question asks about effective treatments or solutions for addressing sudden patchy hair loss and possible causes behind it. The context provides detailed information about various types of hair loss, their causes, and treatments. The AI generated answer correctly identifies that sudden patchy hair loss is known as alopecia areata and lists several treatment options for this condition, including topical corticosteroids, minoxidil, anthralin, immunotherapy, systemic corticosteroids, oral antimalarials, retinoids, or immunosuppressants. The answer also mentions that the cause of alopecia areata is believed to be an autoimmune disorder affecting genetically susceptible individuals exposed to unclear environmental triggers. The information in the AI generated answer is directly derived from the context provided, as it references several treatments mentioned in the context and explains the cause of alopecia areata based on the information presented in the context. Evaluation: Based on the evaluation criteria, I would rate this answer as a 5 because the metric is followed completely. The AI generated answer is derived only from the information presented in the context and accurately reflects the information provided in the context.

Steps to evaluate the context as per the relevance metric: 1. Identify the main aspects of the question: effective treatments or solutions for addressing sudden patchy hair loss and possible causes. 2. Determine if the context provides information on these main aspects. 3. Check if all important aspects are contained in the context and if any irrelevant information is present. The context adheres to the metric considering the question as follows: 1. The context mentions various treatments for sudden patchy hair loss, including topical corticosteroids, minoxidil, anthralin, immunotherapy, oral antimalarials, corticosteroids, retinoids, and immunosuppressants. This directly addresses the question's first aspect (effective treatments or solutions). 2. The context also discusses possible causes of sudden patchy hair loss, such as autoimmune disorders, underlying disorders, and environmental triggers. This addresses the second aspect of the

question. 3. No irrelevant information is present in the context. The metric is followed mostly since all important aspects are addressed, but some additional details could be provided to make the answer more comprehensive (e.g., success rates, side effects, or long-term management strategies). Based on the evaluation criteria: 1 - The metric is not followed at all: 1 2 - The metric is followed only to a limited extent: 2 3 - The metric is followed to a good extent: 3 4 - The metric is followed mostly: 4 5 - The metric is followed completely: 5 Given the information above, I would rate the context as a 4 (followed mostly) on the evaluation criteria.

Query 5: What are the necessary precautions and treatment steps for a person who has fractured their leg during a hiking trip, and what should be considered for their care and recovery?

```
[75]: #print(ground,end="\n\n")
      #print(rel)

      print(textwrap.fill(ground,width=120),end="\n\n")
      print(textwrap.fill(rel,width=120))
```

Steps to evaluate the answer: 1. Identify the main question and the specific information being asked for in the question. 2. Read through the context provided to identify any relevant information related to the question. 3. Determine if the AI generated answer is derived only from the information presented in the context. Explanation: The main question asks about effective treatments or solutions for addressing sudden patchy hair loss and possible causes behind it. The context provides detailed information about various types of hair loss, their causes, and treatments. The AI generated answer correctly identifies that sudden patchy hair loss is known as alopecia areata and lists several treatment options for this condition, including topical corticosteroids, minoxidil, anthralin, immunotherapy, systemic corticosteroids, oral antimalarials, retinoids, or immunosuppressants. The answer also mentions that the cause of alopecia areata is believed to be an autoimmune disorder affecting genetically susceptible individuals exposed to unclear environmental triggers. The information in the AI generated answer is directly derived from the context provided, as it references several treatments mentioned in the context and explains the cause of alopecia areata based on the information presented in the context. Evaluation: Based on the evaluation criteria, I would rate this answer as a 5 because the metric is followed completely. The AI generated answer is derived only from the information presented in the context and accurately reflects the information provided in the context.

Steps to evaluate the context as per the relevance metric: 1. Identify the main aspects of the question: effective treatments or solutions for addressing sudden patchy hair loss and possible causes. 2. Determine if the context provides information on these main aspects. 3. Check if all important aspects are contained in the context and if any irrelevant information is present. The context adheres to the metric considering the question as follows: 1. The context mentions various treatments for sudden patchy hair loss, including topical corticosteroids, minoxidil, anthralin, immunotherapy, oral antimalarials, corticosteroids, retinoids, and immunosuppressants. This directly addresses the question's first aspect (effective treatments or solutions). 2. The context also discusses possible causes of sudden patchy hair loss, such as autoimmune disorders, underlying disorders, and environmental triggers. This addresses the second aspect of the question. 3. No irrelevant information is present in the context. The metric is followed mostly since all important aspects are addressed, but some additional details could be provided to make the answer more comprehensive (e.g., success rates, side effects, or long-term management strategies). Based on the evaluation criteria: 1 - The metric is not followed at all: 1 2 - The metric is followed only to a limited extent: 2 3 - The metric is followed to a good extent: 3 4 - The metric is followed mostly: 4 5 - The metric is followed completely: 5 Given the information above, I would rate the context as a 4 (followed mostly) on the evaluation criteria.

- Following table summarises the observations on Grounding in context and Relevance of the five queries.

```
\begin{array}{ccc} Query&Ground&Relevance\ 1&5&4\ 2&5&4\ 3&5&4\ 4&5&4\ 5&5&4 \\ \end{array}
```

- **Consistent Groundedness:** All five responses appear to be strongly grounded in the text chunks retrieved by the RAG system from the Merck Manual.
- **High Relevance:** The answers consistently and directly address the user queries. They provide the specific type of information requested from the Merck Manual. the context generally contained all important aspects needed to answer the questions and did not include irrelevant information. The rating of 4 (“followed mostly”) suggests that while very good, there was minor room for improvement in the retrieved context itself for the queries.

0.8 Actionable Insights and Business Recommendations

Insights * The system/model demonstrates the capability to ingest an large medical manual and retrieve relevant text. * The responses are overall seen to be adhering to context, factual and repeatable. * There is the possibility of the model providing information outside the context (as seen in the user queries above where pertinent information leaked in from outside the data source - Merck

Manual). * The model is capable of providing accurate & relevant responses for custom queries too (In my case the example of Medial Epicondylitis) * The observations in the “Output Evaluation” section indicates highly grounded and relevant answers retrieved from the Merck Manual. This **RAG model outperforms direct LLM calls**, which were previously observed to provide generic responses. * Increasing the retriever’s k value proved to be valuable in this context. * LLM parameter tuning (non-RAG) showed variable impact. Temperature, top_p, top_k sometimes resulted in identical answers (due to the values chosen). * “repeat_penalty” had a notable impact and introduced some awkward sentences/phrases. * “max_tokens” modifications allowed for more detailed responses. * The information sources need to be updated to reflect FDA guidelines prior to testing in a live medical scenario, but, the progression/direction is encouraging.

Recommendations * Experimentation with larger chunk size is advised considering that 500/1024 characters might be too small in the medical context. * Experimentation with larger overlap is similarly advised. * Consider usage of medical domain specialized embeddings like BioBERT or PubMedBERT. Test thoroughly before switching to a different model. * Usage of medical language needs precision and hence the LLM needs to understand and generate nuanced responses to user queries. Utilization of better engineered prompts could be the key. * Include a disclaimer that conveys that the recommendations of an AI assistant is not a substitute for professional medical judgement. * Manual review/evaluation is essential (by a professional with medical knowledge) for the responses. * If the responses are to be presented for evaluation, then the UI must display the AI generated response along with the source snippets (from the Merck manual, including relevant page numbers/section details) * Given its demonstrated superiority in providing grounded and relevant answers, continued development and refinement of the RAG pipeline should be the primary focus for investment.

By focusing on clinical validation, iterative technical refinement, and a user-centered approach, this RAG-based AI solution can become an invaluable asset in improving healthcare delivery, enhancing diagnostic accuracy, and supporting clinicians in their critical work.

Power Ahead _____